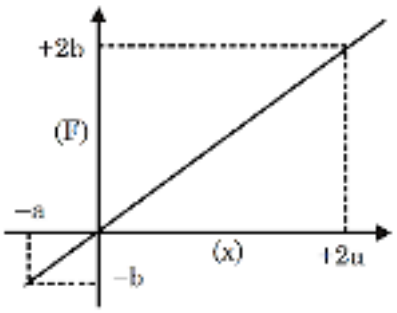
PHYSICS

1) The potential energy of a particle of mass 1kg free to move along x-axis is given by $U(x) = \left(\frac{x^2}{2} - x\right)$ joule. If total mechanical energy of the particle is 2J, then find the maximum speed of the particle.

(1) $\sqrt{5}$m/s
(2) $\sqrt{10}$m/s
(3) $\frac{5}{2}$m/s
(4) 2 m/s

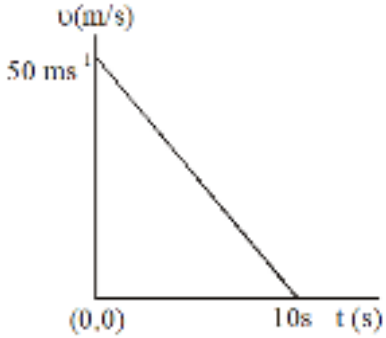
2) A force F acting on a particle varies with the position x as shown in the graph. Find the work done



by the force in displacing the particle from x = –a to x = +2a.

(1) $\frac{3ab}{2}$
(2) $\frac{4ab}{2}$
(3) $\frac{2}{3ab}$
(4) $\frac{2}{4ab}$

3) Velocity-time graph for a body of mass 10 kg is shown in figure. Work-done on the body in first



two seconds of the motion is :-

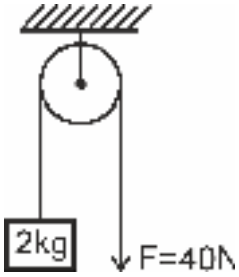
(1) –9300 J

(2) –12000 J

(3) 12000 J

(4) –4500 J

4) A block of mass 2 kg is hanging over a smooth and light pulley through a light string. The other end of the string is pulled by a constant force F = 40 N. At t = 0 the system is at rest as shown. Then in the time interval from t = 0 to t = 2 seconds work done by tension on block will be : (g = 10 m/s^2)



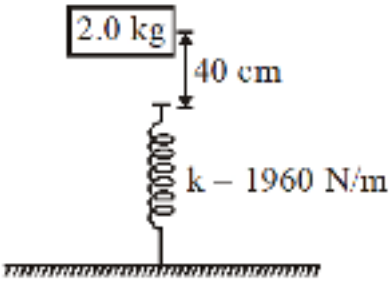
(1) 800 J

(2) 400 J

(3) –400 J

(4) –800 J

5) A 2.0 kg block is dropped from a height of 40 cm onto a spring of spring constant k = 1960 N/m.



Find the maximum distance the spring is compressed :-

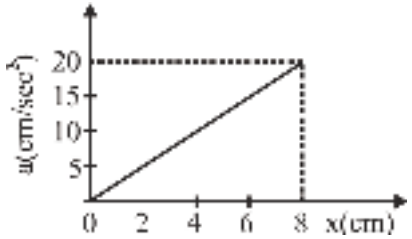
(1) 0.080 m

(2) 0.20 m

(3) 0.40 m

(4) 0.10 m

6) A 10kg mass moves along x-axis. Its acceleration as a function of its position is shown in the figure. What is the total work done on the mass by the force as the mass moves from x = 0 to x = 8 cm



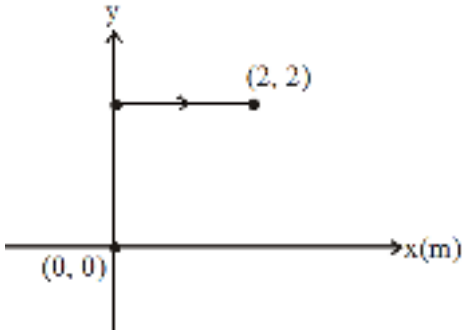
(1) 8×10^{-2} joule

(2) 16×10^{-2} joule

(3) 4×10^{-4} joule

(4) 1.6×10^{-3} joule

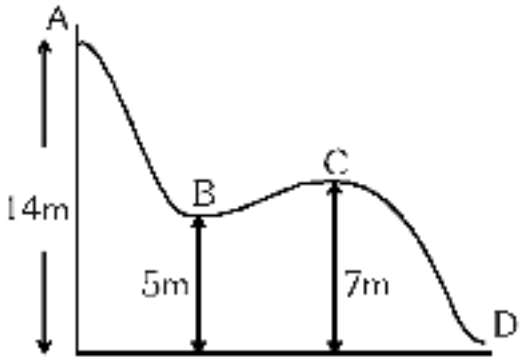
7) Force on a particle is given by $\vec{F} = (2x\hat{i} + 3y\hat{j})$. What is the work done on the particle if it moves from (0, 0) to (2, 2) along the given path.



(1) 3 J
(2) 10 J
(3) 5 J
(4) 4 J

8)

Figure shows the vertical section of frictionless surface. A block of mass 1 kg is released from the position A; its KE as it reaches the position C is (g = 10 m/s^2):-



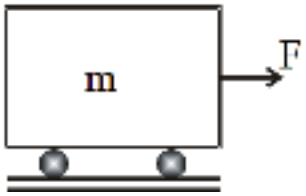
(1) 180J
(2) 140J
(3) 70J
(4) 280J

9) A farmer wants to have delivery rate of water (mass/sec.) three times as that of existing delivery rate from the pump of 2 hp. then he should use the pump of power :

(1) 16 hp
(2) 32 hp
(3) 54 hp
(4) 64 hp

10) A car of mass m is accelerating on a level smooth road under the action of a single force F. The power delivered to the car is constant and equal to P. If the velocity of the car at an instant is v, then

after travelling how much distance it becomes double ?



(1) $\frac{7mv^3}{3P}$

(2) $\frac{4mv^3}{3P}$

(3) $\frac{mv^3}{P}$

(4) $\frac{18mv^3}{7P}$

11) If KE is increased by 400% then % change in momentum :-

(1) 150 %

(2) 123 %

(3) 100 %

(4) 500 %

12) If a force is conservative, then correct statements are :-
(i) work is path independent
(ii) it must be central
(iii) work done round a closed path is zero
(iv) potential energy remains constant

(1) (i,ii)

(2) (ii,iii)

(3) (i,iii)

(4) (i,ii,iv)

13) **Assertion:** The work done by conservative forces does not depend on path taken.
Reason: The work done by non conservative forces depends only on initial and final position.

(1) If both assertion and reason are true and reason is correct explanation of the assertion.

(2) If both assertion and reason are true and reason is not correct explanation of the assertion.

(3) If assertion is true but reason is false.

(4) If assertion and reason both are false.

14) A block of mass m = 2 kg : -

Column I	**Column II**
(P) Lifting the block vertically by 2 m at constant speed (Work by external agent)	(i) 0 J

(Q) A force F = 10 N acts at 60° to horizontal, displacement 4 m	(ii) 40 J
(R) Moving the block 10 m along a smooth horizontal surface	(iii) 20 J
(S) Gravity work when the block slides 5 m down a 30° smooth incline	(iv) 50 J

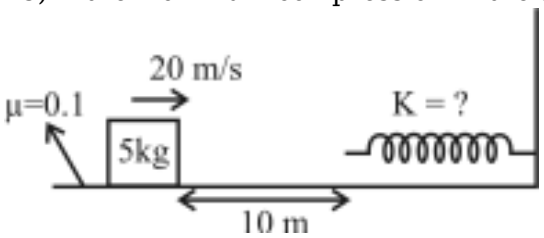
(1) P-(iv), Q-(iii), R-(i), S-(ii)

(2) P-(ii), Q-(iii), R-(i), S-(iv)

(3) P-(ii), Q-(i), R-(iii), S-(iv)

(4) P-(iv), Q-(i), R-(ii), S-(iii)

15) If the maximum compression in the spring is 50 cm then find the value of K(g = 10 m/s^2) :-



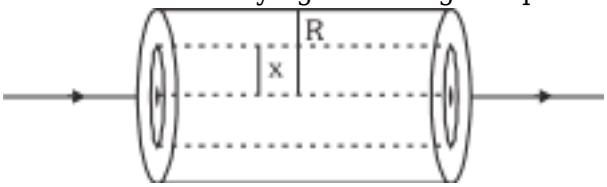
(1) $7580\frac{N}{m}$

(2) $7600\frac{N}{m}$

(3) $8420\frac{N}{m}$

(4) None of these

16) Current density of an area of cross section is varying according to equation $J = J_0\left(1-\frac{x}{R}\right)$. Find the current through the cylinder



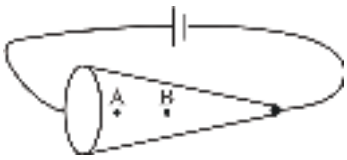
(1) $\pi R^2 J_0$

(2) $\frac{\pi R^2 J_0}{2}$

(3) $\frac{\pi R^2 J_0}{3}$

(4) $\frac{\pi R^2 J_0}{6}$

17) Suppose a current carrying wire has a cross-sectional area that, gradually become smaller along the wire, has the shape of a very long cone as shown in figure. Choose the correct statement:



(1) Electric current is different in different portions of wire.

(2) Electric field at point A is same as that of point B.

(3) Drift speed of electrons at point A is lesser than that of at point B.

(4) Drift speed of electrons at point A is same as that of at point B.

18) Two metallic wires of identical dimensions are connected in parallel. If ρ_1 and ρ_2 are the specific resistances of the metals respectively the effective resistivity of the combination will be :-

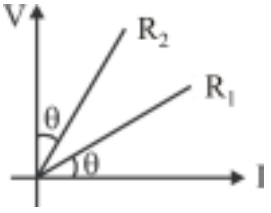
(1) $\frac{\rho_1\rho_2}{\rho_1+\rho_2}$

(2) $\frac{2\rho_1\rho_2}{\rho_1+\rho_2}$

(3) $\frac{\rho_1+\rho_2}{2\rho_1\rho_2}$

(4) $\frac{\rho_1+\rho_2}{\rho_1\rho_2}$

19) The V-I graph for two conductor of resistance R_1 & R_2 are as shown in the figure, the resistance of their series combination $(R_2 + R_1)$ is proportional to :



(1) $\frac{1}{\sin^2\theta\cos^2\theta}$

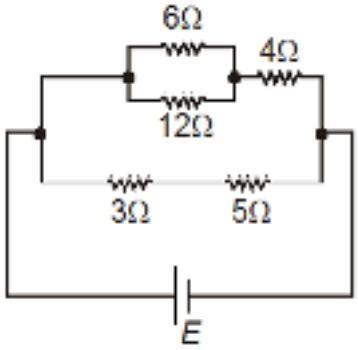
(2) $\frac{1}{\sin\theta\cos\theta}$

(3) $\frac{\cos\theta}{\sin^2\theta}$

(4) $\frac{\tan 2\theta}{\sin^2\theta}$

20)

In the circuit shown, the potential drop across 6 Ω resistor is 12 V. The emf of the ideal battery is



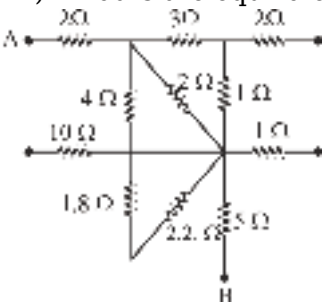
(1) 8 V

(2) 16 V

(3) 24 V

(4) 32 V

21) What is the equivalent resistance between the points A and B of the network :-



(1) $\frac{57}{7}\Omega$

(2) 8 Ω

(3) 6 Ω

(4) $\frac{57}{5}\Omega$

22) Electric field (E) and current density (J) have relation :-

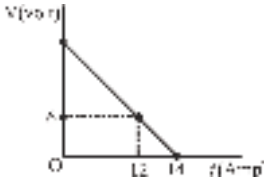
(1) $E \propto J^{-1}$

(2) $E \propto J$

(3) $E \propto \frac{1}{J^2}$

(4) $E^2 \propto \frac{1}{J}$

23) 10 cells, each of emf 'E' and internal resistance 'r', are connected in series to a variable external resistance figure shows the variation of terminal potential difference to their combination with the current drawn from the combination. Emf of each cell is:



(1) 1.6 V

(2) 3.6 V

(3) 1.4 V

(4) 4.2 V

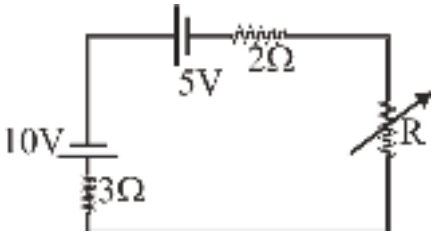
24) To get maximum current through a resistance of 2.5 Ω, one can use m rows of cells, each row having n cells. The internal resistance of each cell is 0.5 Ω. What are the values of n and m if the total number of cells is 45 ?

(1) m= 3, n = 15

(2) m= 5, n = 9

(3) m= 9, n = 5

(4) m= 15, n = 3

25) 

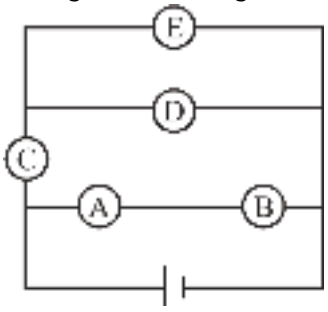
Find maximum power in resistance R :-

(1) 11.25 W

(2) 20 W

(3) 15.5 W

(4) 25 W

26) The given circuit is made up by using identical light bulbs. The light bulbs of maximum brightness of the following will be 

(1) A

(2) C

(3) D

(4) E

27) A galvanometer of 100 Ω resistance yields complete deflection when 10 mA current flows. What should be the value of shunt so that it can measure currents upto 100 mA ?

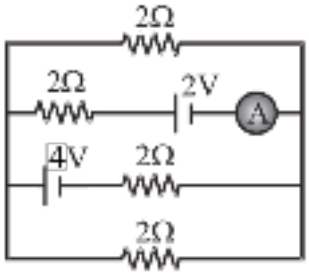
(1) 11.11 Ω

(2) 9.9 Ω

(3) 1.1 Ω

(4) 4.4 Ω

28) The reading of the ammeter as per figure shown is :



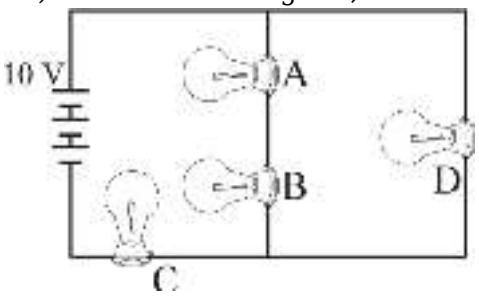
(1) $\frac{1}{8}$A

(2) $\frac{3}{4}$A

(3) $\frac{1}{2}$A

(4) $\frac{1}{4}$A

29) In the circuit diagram, all the bulbs are identical. Which bulb will be the brightest ?



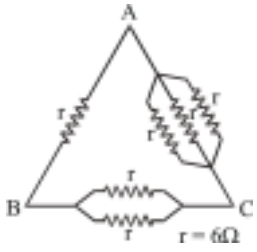
(1) A

(2) B

(3) C

(4) D

30) Six resistances each of value r = 6 Ω are connected between points A, B and C as shown in the figure. If R_1, R_2 and R_3 are the net resistance between A and B, between B and C and between A and C respectively, then $R_1 : R_2 : R_3$ will be equal to :-



(1) 6 : 3 : 2

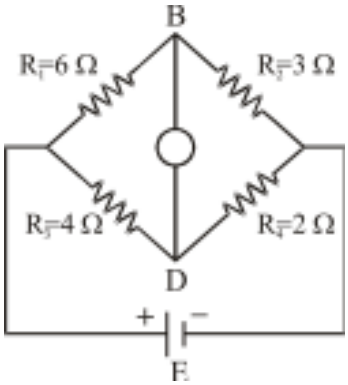
(2) 1 : 2 : 3

(3) 5 : 4 : 3

(4) 4 : 3 : 2

31) Figure shows a balanced wheatstone bridge :

Then which of the following is false ?



(1) There is no flow of current from galvano meter in the balanced condition

(2) If R_1 is slightly increased, the current in galvanometer flows from D to B

(3) If R_4 is slightly decreased, the current in galvanometer flows from D to B

(4) If R_3 is slightly increased, the current in galvanometer flows from B to D

32) A 100μF capacitor is charged to 200 volt. It is discharged through a 2 ohm resistance. The amount of heat generated will be :-

(1) 2 Joule

(2) 4 Joule

(3) 0.2 Joule

(4) 0.4 Joule

33) A parallel plate condenser has a uniform electric field E(V/m) in the space between the plates. If the distance between the plates is d(m) and area of each plate is A(m^2) the energy (joules) stored in the condenser is

(1) $E^2Ad/\in_0$

(2) $\frac{1}{2}\in_0 E^2$

(3) $\in_0 E\ Ad$

(4) $\frac{1}{2}\in_0 E^2\ Ad$

34) On placing a dielectric slab between the plates of a charged condenser its:-

	Capacity	**Charge**	**Potential Difference**	**Energy stored**
(1)	decreases	remains unchanged	decreases	increases
(2)	increases	remains unchanged	increases	increases
(3)	increases	remains unchanged	decreases	decreases

(4)	decreases	remains unchanged	increases	decreases

(1) 1

(2) 2

(3) 3

(4) 4

35) Voltage rating of a parallel plate capacitor is 500V. Its dielectric can withstand a maximum electric field of 10^6 V/m. The plate area is 10^{-4} m^2. What is the dielectric constant is the capacitance is 15 pF?
(given $\epsilon_0 = 8.86 \times 10^{-12}$ C^2/Nm^2)

(1) 3.8

(2) 4.5

(3) 6.2

(4) 8.5

36) A $600\,\mu F$ capacitor is charged by a 200 V supply. It is then disconnected from the supply and is connected to another uncharged $600\,\mu F$ capacitor. How much electrostatic energy is lost in the process ?

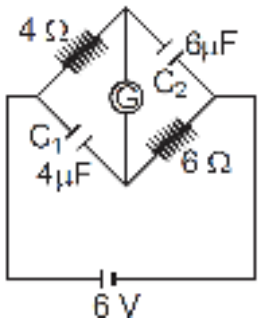
(1) 3 J

(2) 6 J

(3) 9 J

(4) 12 J

37) A galvanometer (G) of 2Ω resistance is connected in the given circuit. The ratio of charge stored in C_1 and C_2 is :



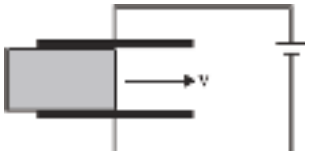
(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

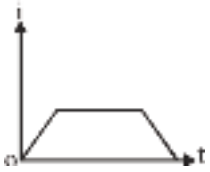
(3) 1

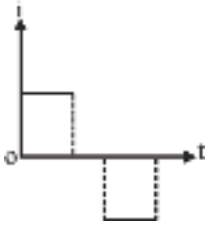
(4) $\frac{1}{2}$

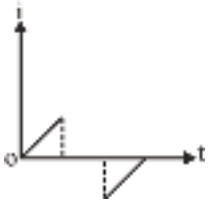
38) A dielectric slab of area A is inserted between the plates of a capacitor of area 2A with constant speed v as shown in figure. Distance between the plates is d.

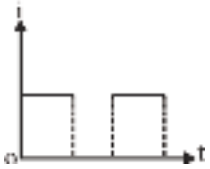


The capacitor is connected to a battery of emf E. The current in the circuit varies with time as

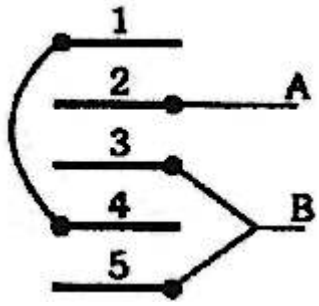
(1) 

(2) 

(3) 

(4) 

39) Five identical metal plates 1, 2, 3, 4 and 5 each of area A on one side are fixed parallel and equidistant (d) to each other. The plates 1 and 4 are joined by a conductor and plates 3 and 5 are also joined by a conductor as shown in figure. Then, the capacitance of this system between A and B is :



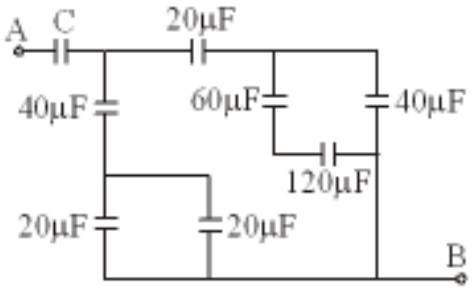
(1) $\frac{5\varepsilon_0 A}{d}$

(2) $\frac{4\varepsilon_0 A}{d}$

(3) $\frac{5\varepsilon_0 A}{3d}$

(4) None

40) Figure shows a network of capacitors. Find the value of C, if the effective capacitance between



the points A and B is 12 μF.

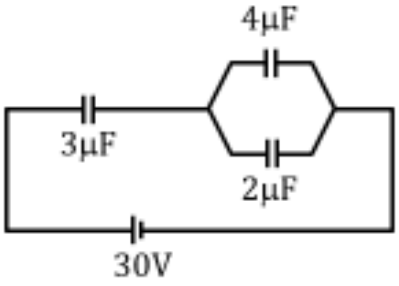
(1) 18 μF

(2) 20 μF

(3) 10 μF

(4) 5 μF

41) In the circuit shown in figure, match the following two columns.



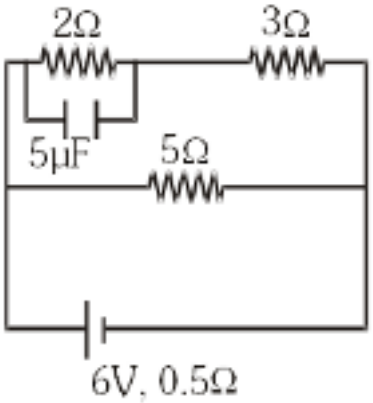
Column-I	**Column-II**
(I) Charge given by battery	(a) 20 μC
(II) Charge of 4μF capacitor	(b) 30 μC
(III) Charge of 2 μF capacitor	(c) 60 μC
(IV) Charge of 3 μF capacitor	(d) 40 μC

(1) I → a ; II → b ; III → b ; IV → c

(2) I → c ; II → d ; III → a ; IV → c

(3) I → b ; II → a ; III → c ; IV → d

(4) I → c ; II → d ; III → c ; IV → a



42) Find out amount of charge on capacitor shown in circuit.

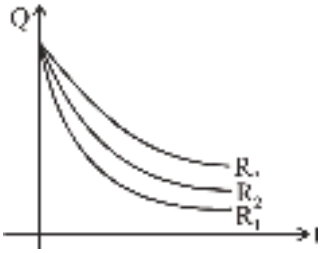
(1) 0μC

(2) 30μC

(3) 10μC

(4) 25μC

43) Three identical capacitors are given a charge Q each and they are then allowed to discharge through resistance R_1, R_2 and R_3. Their charges as a function of time shown in the graph below. The smallest of the three resistance is :-



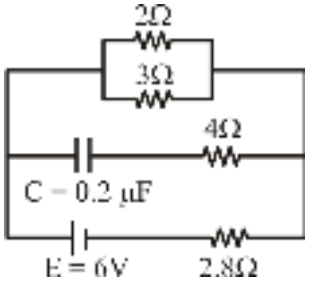
(1) R_3

(2) R_2

(3) R_1

(4) Cannot be predicted

44) In the circuit shown, the internal resistance of the cell is negligible. The steady state current in the 2 Ω resistor is :-



(1) 0.6 A

(2) 0.9 A

(3) 1.2 A

(4) 1.5 A

45) Two parallel plate capacitors whose capacitances are C and 2 C respectively, are joined in parallel. These are charged to V potential difference. If the battery is now removed and a dielectric of dielectric constant K is filled in between the plates of the capacitor 2 C, then what will be the potential difference across each capacitor ?

(1) $\frac{V}{K+2}$

(2) $\frac{2V}{K+2}$

(3) $\frac{3V}{K+2}$

(4) $\frac{3V}{(2K+1)}$

CHEMISTRY

1) E.M.F. of the cell
$2Ag^{+} + Cu \rightarrow Cu^{+2} + 2Ag$
[Given : $E^{o}_{Ag|Ag^{+}} = -0.8$ V ; $E^{o}_{Cu^{+2}|Cu} = 0.3$ V]

(1) –0.5 V
(2) 0.5 V
(3) –1.1 V
(4) 1.1 V

2) Sn^{4+} (aq) + $2e^{-} \rightarrow Sn^{2+}$ (aq) E°=0.15V
Cr^{3+} (aq)+ $e^{-} \rightarrow Cr^{2+}$ (aq) E°=–0.41V
According to the standard reduction potentials above, what is the value of E° for the reaction below?
$2Cr^{3+}$ (aq) + Sn^{2+} (aq) → $2Cr^{2+}$ (aq) +Sn^{4+}(aq)

(1) –0.97 V
(2) –0.56 V
(3) +0.56 V
(4) +0.97 V

3) Given : $E^{o}_{Cr^{3+}/Cr} = -0.72V, E^{o}_{Fe^{2+}/Fe} = -0.42V$
The potential for the cell
Cr | Cr^{3+} (0.1 M) || Fe^{2+} (0.01 M) | Fe, is:

(1) 0.339 V
(2) –0.339 V
(3) –0.26 V
(4) 0.26 V

4) The electrode potential becomes equal to standard electrode potential when reactants and products concentration ratio is:

(1) Equal to 1
(2) Greater than 1
(3) Less than 1
(4) None of these

5) The emf of the cell
Zn | Zn^{2+} (0.01 M) || Fe^{2+} (0.001 M) | Fe
at 298 K is 0.2905 then the value of equilibrium constant for the cell reaction is:

(1) $\frac{0.32}{e^{0.0295}}$
(2) $10^{\frac{0.32}{0.0295}}$

(3) $10^{\frac{0.26}{0.0591}}$

(4) $10^{\frac{0.32}{0.0591}}$

6) The molar conductivities $\wedge^0_{NaOAc}$ and $\wedge^0_{HCl}$ at infinite dilution in water at 25°C are 91.0 and 426.2 S cm^2 mol^{-1} respectively. To calculate $\wedge^0_{HOAc}$ the additional value required is :

(1) $\wedge^0_{NaCl}$

(2) $\wedge^0_{H_2O}$

(3) $\wedge^0_{KCl}$

(4) $\wedge^0_{NaOH}$

7) How many faradays are required to reduce one mole of MnO_4^- to Mn^{2+} ?

(1) 1

(2) 2

(3) 3

(4) 5

8) Three faradays of electricity was passed through an aqueous solution of iron (II) bromide. The mass of iron metal (at. mass = 56) deposited at the cathode is :

(1) 56 g

(2) 84 g

(3) 112 g

(4) 168 g

9) How many electrons flow when a current of 5 amperes is passed through a conductor for 200 second ?

(1) 6.214×10^{21}

(2) 60.241×10^{21}

(3) 6.241×10^{22}

(4) 6.0241×10^{22}

10) The specific conductivity of a saturated solution AgCl is 3.40×10^{-6} $ohm^{-1}cm^{-1}$ at 25° C. If molar conductance of Ag^+ = 62.3 ohm^{-1} cm^2 mol^{-1} and molar conductance of Cl^- = 67.7 ohm^{-1} cm^2 mol^{-1} at infinite dilution, the solubility of AgCl at 25° C is:

(1) 2.6×10^{-5} M

(2) 4.5×10^{-3} M

(3) 3.6×10^{-5} M

(4) 3.6×10^{-3} M

11) The common features among the species CN^-, CO and NO^+ are :

(1) bond order three and diamagnetic

(2) bond order three and weak field ligands

(3) paramagnetic and strong field ligands

(4) paramagnetic and π-acceptor ligands

12) Which of the following ambidentate ligand can act as flexidentate ligand ?

(1) SCN^-

(2) OCN^-

(3) SO_3^{2-}

(4) None of these

13) CO bond length is longest in :

(1) $[Mn(CO)_6]^+$

(2) $[V(CO)_6]^-$

(3) $Cr(CO)_6$

(4) $[Ti(CO)_6]^{2-}$

14) What is the oxidation number of Fe in $[Fe(H_2O)_5(NO)]^{2+}$ ion?

(1) +2

(2) +3

(3) +1

(4) 0

15) Incorrect match for given complex compound/ion and its characteristics :

(1) $[CrBrCl(en)_2]Br$ → Ionization and optical isomerism

(2) $[CoBr_3(H_2O)_3]$ → Fac-mer and hydrate isomerism

(3) $[PtCl_2(NH_3)_4]$ $[Co(SCN)_4]$ → Linkage isomerism and paramagnetic character

(4) $[Co(ox)_3]^{3-}$ → Inner orbital complex and optical isomerism

16) Find IUPAC name of the hydrate isomer of $CrCl_3.6H_2O$, which is having lowest electrical conductivity.

(1) Hexaaquachromium(III)chloride

(2) Tetraaquadichloridochromium(III)chloridedihydrate

(3) Pentaaquachloridochromium(III)chloridemonohydrate

(4) Triaquatrichloridochromium(III)trihydrate

17) If EAN of central metal cation M^{2+} in a non-chelating complex is 36 and atomic no. of metal M is 26, then the number of monodentate ligand in this complex are :

(1) 5

(2) 4

(3) 6

(4) None of these

18) Which of the following pair the EAN of central metal atom is not same ?

(1) $[Fe(CN)_6]^{3-}$ and $[Fe(NH_3)_6]^{3+}$

(2) $[Cr(NH_3)_6]^{3+}$ and $[Cr(CN)_6]^{3-}$

(3) $[FeF_6]^{3-}$ and $[Fe(CN)_6]^{3-}$

(4) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$

19) IUPAC name of $[Fe(O_2)(CN)_4Cl]^{4-}$ is :

(1) chlorotetracyano dioxoferrate (IV)ion

(2) chlorotetracyano peroxoferrate (II)ion

(3) chlorotetracyano superoxoferrate (II)ion

(4) tetracyanochloro superoxoferrate (II)ion

20) Select the correct IUPAC name of Prussian blue :

(1) Iron(II) hexacyanoferrate(II)

(2) Iron(II) hexacyanoferrate(III)

(3) Iron(III) hexacyanoferrate(II)

(4) Iron(III) hexacyanoferrate(III)

21) Which of the following complex compound is “Pentaaquacyanidoiron (III)trichloridotricyanido cobaltate (III)” ?

(1) $[Fe(CN)(H_2O)_5][CoCl_3(CN)_3]$

(2) $[Fe(CN)(H_2O)_5]_2[CoCl_3(CN)_3]_3$

(3) $[Fe(CN)_2(H_2O)_4]_3[FeCl_3(CN)_3]_2$

(4) $[Fe(CN)(H_2O)_5]_3[CoCl_3(CN)_3]_2$

22) A platinum complex of ammonia and chlorine produces four ions per molecule in the solution. It is:

(1) $[Pt(NH_3)_5Cl]Cl_3$

(2) $[Pt(NH_3)_6]Cl_4$

(3) $[Pt(NH_3)_2Cl_4]$

(4) $[Pt(NH_3)_4Cl_2]Cl_2$

23) Iron carbonyl, $Fe(CO)_5$ is :

(1) tetranuclear

(2) mononuclear

(3) trinuclear

(4) dinuclear

24) What is the correct electronic configuration of the central atom in $K_4[Fe(CN)_6]$ based on crystal field theory ?

(1) $e^4t_2^2$

(2) $t_2^6e^0$

(3) $t_{2g}^6e_g^0$

(4) $e^3t_2^3$

25) Which of the following is the correct order of increasing field strength of ligands to form coordination compounds ?

(1) $SCN^- < F^- < CN^- < C_2O_4^{2-}$

(2) $F^- < SCN^- < C_2O_4^{2-} < CN^-$

(3) $CN^- < C_2O_4^{2-} < SCN^- < F^-$

(4) $SCN^- < F^- < C_2O_4^{2-} < CN^-$

26) Match List-I with List-II.

List-I		**List-II**	
(A)	$[Fe(CN)_6]^{3-}$	(i)	5.92 BM
(B)	$[Fe(H_2O)_6]^{3+}$	(ii)	0 BM
(C)	$[Fe(CN)_6]^{4-}$	(iii)	4.90 BM
(D)	$[Fe(H_2O)_6]^{2+}$	(iv)	1.73 BM

Select the **correct** answer from the options given below.

(1) (A)-(iv), (B)-(ii), (C)-(i), (D)-(iii)

(2) (A)-(ii), (B)-(iv), (C)-(iii), (D)-(i)

(3) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

(4) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

27) Which of the following complex ions is expected to absorb visible light ?

(1) $[Ti(en)_2(NH_3)_2]^{4+}$

(2) $[Cr(NH_3)_6]^{3+}$

(3) $[Zn(NH_3)_6]^{2+}$

(4) $[Sc(H_2O)_3(NH_3)_3]^{3+}$
[At. no. : Zn = 30, Sc = 21, Ti = 22, Cr = 24]

28) Which one of the following has an optical isomer ?

(1) $[Co(H_2O)_4(en)]^{3+}$

(2) $[Zn(en)_2]^{2+}$

(3) $[Zn(en)(NH_3)_2]^{2+}$

(4) $[Co(en)_3]^{3+}$

29) Which of the following complex has minimum magnitude of Δ_0?

(1) $[Cr(CN)_6]^{3-}$

(2) $[Co(NH_3)_6]^{3+}$

(3) $[CoCl_6]^{3-}$

(4) $[Cr(H_2O)_6]^{3+}$

30) The ligands in anti-cancer drug cis platin are :

(1) NH_3, Cl

(2) NH_2^-, O_2^{-2}

(3) Cl, H_2O

(4) NO, Cl

31) Which of the following is π acid ligand ?

(1) NH_3

(2) CO

(3) F^-

(4) Ethylene diamine

32) **Statement-1 :** In N_2H_4, any one N-atom can coordination with central metal cation in a coordination compound.
Statement-2 : N_2H_4 can also act as ambidentate ligand.

(1) If statement-1 is true but the statement-2 is false.

(2) If statement-1 is false but statement-2 is true.

(3) If both statements are true and the statement-2 is the correct explanation of statement-1.

(4) If both statements are true and the statement-2 is not the correct explanation of statement-1.

33) **Statement-1 :** $[Ti(H_2O)_6]^{4+}$ is coloured while $[Sc(H_2O)_6]^{3+}$ is colourless.
Statement-2 : d-d transition is not possible in $[Sc(H_2O)_6]^{3+}$ complex.

(1) If statement-1 is true but the statement-2 is false.

(2) If statement-1 is false but statement-2 is true.

(3) If both statements are true and the statement-2 is the correct explanation of statement-1.

(4) If both statements are true and the statement-2 is not the correct explanation of statement-1.

34) **Statement-1 :** Zeise’s salt is π-bonded organometallic compound.

Statement-2 : Zeise's salt contain C_2H_4 molecules as one of the ligand.

(1) If statement-1 is true but the statement-2 is false.

(2) If statement-1 is false but statement-2 is true.

(3) If both statements are true and the statement-2 is the correct explanation of statement-1.

(4) If both statements are true and the statement-2 is not the correct explanation of statement-1.

35) **Statement-1 :** $[CoF_6]^{3-}$ is paramagnetic.
Statement-2 : Co^{3+} has $3d^6$ outer electronic configuration. The unpaired electrons do not pair up because of weak field provided by F^-.

(1) If statement-1 is true but the statement-2 is false.

(2) If statement-1 is false but statement-2 is true.

(3) If both statements are true and the statement-2 is the correct explanation of statement-1.

(4) If both statements are true and the statement-2 is not the correct explanation of statement-1.

36) The reduction potential values are given below :
Al^{3+}/Al = –1. 67 volt, Mg^{2+}/Mg = –2.34 volt,
Cu^{2+}/Cu = +0.34 volt, $I_2/2I^-$ = +0.53 volt.
Which one is the best reducing agent ?

(1) Al

(2) Mg

(3) Cu

(4) I_2

37) During the electrolysis of fused NaCl, which reaction occurs at anode ?

(1) Chloride ions are oxidised

(2) Sodium ions are oxidised

(3) Chloride ions are reduced

(4) Sodium ions are reduced

38) The amount of electricity required to produce one mole of copper from copper sulphate solution will be :

(1) 1 faraday

(2) 2.33 faraday

(3) 2 faraday

(4) 1.33 faraday

39) The resistance of 0.01 N solution of an electrolyte was found to be 210 ohm at 298 K using a conductivity cell of cell constant 0.66 cm^{-1}. The equivalent conductance of solution is :–

(1) 314.28 mho cm^2 eq^{-1}

(2) 3.14 mho cm^2 eq^{-1}

(3) 314.28 mho^{-1} cm^2 eq^{-1}

(4) 3.14 mho^{-1} cm^2 eq^{-1}

40) The specific conductance of a N/10 solution of KCl at 25°C is 0.0035 Ω^{-1} cm^{-1}. If the resistance of a cell containing this solution is 500 Ω, what is the cell constant?

(1) 1.106 cm^{-1}

(2) 1.75 cm^{-1}

(3) 3.2 cm^{-1}

(4) 2 cm^{-1}

41) What is the value of ΔG° for the reaction?
$2A1(s) + 3Cu^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Cu(s)$ $E° = 2.02$ V

(1) –1170kJ

(2) –585kJ

(3) –390kJ

(4) –195kJ

42) Three mole of electrons are passed through three solution in succession containing $AgNO_3$, $CuSO_4$ and $AuCl_3$ respectively.
The ratio of moles of cations reduced at cathode will be :

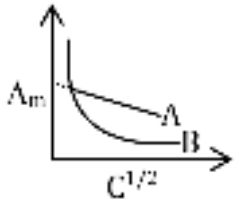
(1) 1 : 2 : 3

(2) 2 : 1 : 3

(3) 3 : 2 : 1

(4) 6 : 3 : 2

43)



A and B respectively are :-

(1) HCN, NaCl

(2) KCl, NaCl

(3) KCl, CH_3COOH

(4) CH_3COOH, HCN

44) Equivalent conductivity of $Fe_2(SO_4)_3$ is related to molar conductivity by the expression :-

(1) $\Lambda_{eq} = 3\Lambda_m$

(2) $\Lambda_{eq} = \frac{\Lambda_m}{6}$

(3) $\Lambda_{eq} = \Lambda_m$

(4) $\Lambda_{eq} = \frac{\Lambda_m}{3}$

45) Saturated solution of KNO_3 is used to make 'salt-bridge' because :

(1) velocity of both K^+ and NO_3^- are nearly the same
(2) velocity of NO_3^- is greater than that of K^+
(3) velocity of K^+ is greater than that of NO_3^-
(4) KNO_3 is high soluble in water

BIOLOGY

1) Which of the following statements regarding the human testis is **correct**?

(1) Each testicular lobule contains 1–3 highly coiled seminiferous tubules.
(2) Seminiferous tubules are lined only by columnar epithelial cells.
(3) Leydig cells are present inside the lumen of seminiferous tubules.
(4) Sertoli cells secrete testosterone.

2) During spermatogenesis, which sequence correctly represents the developmental stages?

(1) Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatid → Spermatozoa
(2) Spermatogonia → Secondary spermatocyte → Primary spermatocyte → Spermatid → Spermatozoa
(3) Primary spermatocyte → Spermatogonia → Spermatid → Secondary spermatocyte → Spermatozoa
(4) Spermatogonia → Spermatid → Primary spermatocyte → Secondary spermatocyte → Spermatozoa

3) A primary spermatocyte in humans contains **46 chromosomes and 92 chromatids**.
After completion of meiosis I, each secondary spermatocyte will contain:

(1) 46 chromosomes and 46 chromatids
(2) 23 chromosomes and 46 chromatids
(3) 23 chromosomes and 23 chromatids
(4) 46 chromosomes and 23 chromatids

4) Which of the following changes occurs during spermiogenesis?

(1) Formation of spermatids by meiosis II
(2) Transformation of spermatids into spermatozoa
(3) Formation of primary spermatocytes
(4) Formation of secondary spermatocytes

5) Which hormone primarily stimulates the **Leydig cells** of the testes?

(1) FSH
(2) Prolactin
(3) LH
(4) Oxytocin

6) Which of the following is not a function associated with Sertoli cells?

(1) Providing nutrition to developing germ cells
(2) Supporting spermatogenesis
(3) Secretion of inhibin
(4) Secretion of testosterone

7) Which statement regarding human oogenesis is correct?

(1) Primary oocytes are formed only after puberty.
(2) Primary oocytes remain arrested in prophase I until puberty.
(3) Secondary oocyte is diploid.
(4) Oogonia continue to multiply throughout reproductive life.

8) In a normal 28-day menstrual cycle, ovulation generally occurs around:

(1) Day 5
(2) Day 10
(3) Day 14
(4) Day 21

9) Which event is most directly responsible for triggering ovulation?

(1) Sudden fall in FSH
(2) LH surge
(3) Increase in progesterone
(4) Degeneration of corpus luteum

10) Which combination is correctly matched?

(1) Menstrual phase — maximum progesterone secretion
(2) Luteal phase — shedding of endometrium
(3) Ovulatory phase — degeneration of corpus luteum
(4) Follicular phase — development of ovarian follicles

11) The corpus luteum primarily secretes:

(1) FSH
(2) LH
(3) Progesterone
(4) hCG

12) **Assertion (A):** Progesterone levels decrease sharply if fertilisation does not occur.
Reason (R): The corpus luteum degenerates in the absence of pregnancy.

(1) Both A and R are true, and R correctly explains A
(2) Both A and R are true, but R does not explain A
(3) A is true, R is false
(4) A is false, R is true

13) In humans, fertilisation normally occurs in the:

(1) Uterus
(2) Cervix
(3) Ampullary-isthmic junction of the fallopian tube
(4) Vagina

14) Which structure of the sperm is primarily involved in penetration of the zona pellucida?

(1) Middle piece
(2) Acrosome
(3) Mitochondrial sheath
(4) Centriole

15) After fusion of sperm and oocyte membranes, which event prevents entry of additional sperms?

(1) Acrosome reaction
(2) Cortical reaction
(3) Capacitation
(4) Spermiation

16) At the time of ovulation, the human female releases:

(1) Primary oocyte
(2) Secondary oocyte arrested at metaphase II
(3) Ovum arrested at metaphase I
(4) Mature ovum after completion of meiosis II

17) Which sequence correctly represents early embryonic development?

(1) Zygote → Morula → Blastocyst → Implantation
(2) Zygote → Blastocyst → Morula → Implantation
(3) Morula → Zygote → Blastocyst → Implantation
(4) Zygote → Implantation → Morula → Blastocyst

18) Implantation of the blastocyst normally occurs in the:

(1) Myometrium

(2) Endometrium
(3) Perimetrium
(4) Cervical canal

19) The outer layer of cells of the blastocyst that contributes to the formation of the placenta is:

(1) Inner cell mass
(2) Trophoblast
(3) Amnion
(4) Embryoblast

20) Which hormone is detected in maternal blood and urine during early pregnancy and is commonly used for pregnancy testing?

(1) Progesterone
(2) Estrogen
(3) hCG
(4) FSH

21) Which of the following hormones is not secreted by the placenta?

(1) hCG
(2) hPL
(3) Estrogen
(4) FSH

22) During early pregnancy, hCG maintains the:

(1) Graafian follicle
(2) Corpus luteum
(3) Primary oocyte
(4) Seminiferous tubule

23) Which combination is most directly associated with the initiation and progression of parturition?

(1) Oxytocin and cortisol
(2) FSH and inhibin
(3) hCG and FSH
(4) Testosterone and LH

24) Which statement regarding lactation is correct?

(1) Prolactin causes milk ejection.
(2) Oxytocin causes milk production.
(3) hCG causes milk ejection.
(4) Prolactin stimulates milk production, while oxytocin causes milk ejection.

25) Read the following statements:
A. LH acts on Leydig cells and stimulates androgen secretion.
B. FSH acts on Sertoli cells and supports spermatogenesis.
C. Fertilisation normally occurs at the ampullary-isthmic junction.
D. Implantation occurs in the myometrium.
E. hCG maintains the corpus luteum during early pregnancy.
Which of the following represents the **correct statements**?

(1) A, B and C only
(2) A, B, C and E only
(3) B, C and D only
(4) A, C, D and E only

26) Which of the following contraceptive methods prevents ovulation and also causes changes in cervical mucus?

(1) Copper-T
(2) Vasectomy
(3) Combined oral contraceptive pills
(4) Condoms

27) Which of the following statements regarding **Cu-T** is correct?

(1) It prevents ovulation in all users.
(2) It increases sperm motility.
(3) It increases phagocytosis of sperms within the uterus.
(4) It permanently blocks the fallopian tubes.

28) Which of the following is a hormone-releasing IUD?

(1) Cu-T
(2) Multiload 375
(3) LNG-20
(4) Lippes loop

29) **Assertion (A):** Condoms are considered a useful method for preventing both pregnancy and sexually transmitted infections.
Reason (R): Condoms act as a physical barrier between the reproductive tracts of sexual partners.

(1) Both A and R are true, and R correctly explains A
(2) Both A and R are true, but R does not explain A
(3) A is true, but R is false
(4) A is false, but R is true

30) In vasectomy, which structure is cut and tied?

(1) Ureter

(2) Vas deferens
(3) Epididymis
(4) Seminiferous tubule

31) Tubectomy prevents pregnancy primarily by:

(1) Preventing ovulation
(2) Preventing implantation
(3) Preventing transport of sperm and ovum through the fallopian tube
(4) Destroying the corpus luteum

32) Emergency contraceptive pills are most effective when taken:

(1) Within 24 hours only
(2) Within 72 hours of unprotected intercourse
(3) After implantation
(4) Only after a positive pregnancy test

33) Which of the following is not a natural method of contraception?

(1) Periodic abstinence
(2) Withdrawal method
(3) Lactational amenorrhea
(4) Cu-T

34) Lactational amenorrhea is effective as a contraceptive method mainly because:

(1) Ovulation is temporarily suppressed during intense lactation.
(2) Fertilisation becomes impossible permanently.
(3) Fallopian tubes become blocked.
(4) It destroys sperm in the uterus.

35) Which of the following is not primarily transmitted through sexual contact?

(1) Gonorrhoea
(2) Syphilis
(3) Malaria
(4) Genital herpes

36) Which pair is **incorrectly matched**?

(1) Gonorrhoea — *Neisseria gonorrhoeae*
(2) Syphilis — *Treponema pallidum*
(3) Genital herpes — Herpes simplex virus
(4) AIDS — *Plasmodium*

37) Which of the following may be a symptom of sexually transmitted infections?

(1) Genital itching
(2) Abnormal genital discharge
(3) Painful urination
(4) All of the above

38) Which of the following is not an appropriate strategy for preventing sexually transmitted infections?

(1) Avoiding sexual contact with unknown/infected partners
(2) Using condoms
(3) Maintaining sexual hygiene
(4) Sharing needles with an infected person

39) Medical Termination of Pregnancy (MTP) is considered relatively safer when performed:

(1) During the first trimester
(2) Only during the third trimester
(3) After completion of pregnancy
(4) Only after implantation is complete

40) A couple is unable to conceive because the woman's fallopian tubes are blocked. Which ART would be particularly useful?

(1) IVF followed by embryo transfer
(2) Vasectomy
(3) Tubectomy
(4) Lactational amenorrhea

41) GIFT involves:

(1) Transfer of a zygote into the uterus
(2) Transfer of gametes into the fallopian tube
(3) Transfer of an embryo into the uterus only
(4) Injection of sperm directly into the ovum

42) Which ART involves the direct injection of a sperm into the ovum?

(1) IVF-ET
(2) GIFT
(3) ICSI
(4) ZIFT

43) ZIFT involves transfer of:

(1) Gametes into the fallopian tube

(2) Zygote/early embryo into the fallopian tube

(3) Blastocyst into the ovary

(4) Sperm into the uterus

44) Match List I with List II:

List I	List II
A. GIFT	P. Sperm injected directly into ovum
B. ICSI	Q. Gametes transferred into fallopian tube
C. ZIFT	R. Zygote/early embryo transferred into fallopian tube
D. IVF-ET	S. Embryo transferred into female reproductive tract

(1) A-R, B-S, C-P, D-Q

(2) A-P, B-Q, C-S, D-R

(3) A-Q, B-P, C-R, D-S

(4) A-S, B-R, C-Q, D-P

45) Read the following statements:
A. Cu-T increases phagocytosis of sperms in the uterus.
B. Hormone-releasing IUDs make the uterus unsuitable for implantation and the cervix hostile to sperms.
C. Vasectomy prevents sperm formation in the testes.
D. Condoms can help prevent STIs.
E. ICSI involves direct injection of sperm into the ovum.
Which statements are correct?

(1) A, C and D only

(2) B, C, D and E only

(3) A, B, D and E only

(4) A, B, C, D and E

46) Choose the mis-match :

(1) Amylose → contains α-1, 4 glycosidic bond.

(2) K^+ → most abundant mineral element in ICF.

(3) Na^+ → most abundant mineral element in ECF.

(4) Cellulose → violet colour with iodine solution.

47) Which of the following is not strictly a biomacromolecule?

(1) Proteins

(2) Lipids

(3) Polysaccharides

(4) Nucleic acid

48) An alpha helix represents

(1) primary structure of protein
(2) aggregation of protein
(3) secondary structure of protein
(4) tertiary structure of protein

49) The full name of NAD is :-

(1) Nicotinamide adenosine dinucleotide
(2) Nicotinamide adenine dinucleotide
(3) Nicotinamide adenine diamide
(4) Nicotinamide adenosine diamide

50) With continuously increasing substrate concentration enzymatic activity :-

(1) Continuously increases
(2) Continuously decreases
(3) First increases then decreases
(4) First increases then become constant

51) Substrate concentration at which an enzyme attains half of its maximum velocity is?

(1) Half-life of enzyme
(2) Km-constant of enzyme
(3) Concentration ratio
(4) Turn over number (TON) of enzyme E

52) Which of the following statements about polysaccharides is correct?

(1) Cellulose is a polymer of fructose and forms helical structures that bind iodine to give a blue color.
(2) Starch forms helical structures and gives a blue color when treated with iodine solution.
(3) Glycogen is a homopolymer of amino sugars found in plant cell walls.
(4) Chitin is a polymer of glucose and is found in the human liver.

53) Proteins present in protoplasm are very important because:-

(1) They provide definite shape to cell.
(2) They function as biocatalyst.
(3) They yield energy.
(4) They are stored food.

54) Which of the following amino acids is correctly matched with its R group?

(1) Glycine – Methyl group
(2) Alanine – Hydroxymethyl group
(3) Serine – Hydroxymethyl group

(4) Valine – Aromatic ring

55) Adenylic acid is: -

(1) Ribose + Adenosine + phosphate
(2) Ribose + cytosine
(3) Nucleoside
(4) Deoxyribonucleotide

56) Allosteric enzymes have allosteric sites for

(1) Inhibition only
(2) Activation only
(3) Reduction in activation energy
(4) Both activation and inhibition

57) Enzyme inhibition caused by a product of the same biochemical reaction is :-

(1) Competitive inhibition
(2) Feed-back inhibition
(3) In competitive inhibition
(4) Semi-competitive inhibition

58) Which of the following statements regarding enzyme inhibition is correct ?

(1) Non-competitive inhibition of an enzyme can be overcome by adding large amount of substrate
(2) Competitive inhibition is seen, when a substrate competes with an enzyme for binding to an inhibitor protein
(3) Competitive inhibition is seen, when the substrate and the inhibitor compete for the active site on the enzyme
(4) Non-competitive inhibitors often bind to the enzyme irreversibly.

59) Prosthetic groups differ from co-enzymes in that :-

(1) they require metal ions for their activity.
(2) they are tightly bound to apoenzymes.
(3) their association with apoenzymes is transient.
(4) they can serve as co-factors in a number of enzyme-catalyzed reactions.

60) Following are the statements with reference to 'lipids'.
(a) Lipids having only single bonds are called unsaturated fatty acids.
(b) Lecithin is a phospholipid.
(c) Trihydroxy propane is glycerol.
(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
(e) Arachidonic acid has 16 carbon atoms.
Choose the **correct** answer from the options given below.

(1) (a) and (b) only

(2) (c) and (d) only

(3) (b) and (c) only

(4) (b) and (e) only

61) Read the following statements on lipids and find out correct set of statements:
(a) Lecithin found in the plasma membrane is a glycolipid.
(b) Saturated fatty acids possess one or more C = C bonds.
(c) Gingelly oil has lower melting point, hence remains as oil in winter.
(d) Lipids are generally insoluble in water but soluble in some organic solvents.
(e) When fatty acid is esterified with glycerol, monoglycerides are formed.
Choose the correct answer from the options given below:

(1) (a), (d) and (e) only

(2) (c), (d) and (e) only

(3) (a), (b) and (d) only

(4) (a), (b) and (c) only

62) Regarding catalytic cycle of an enzyme action, select the correct sequential steps :
A. Substrate enzyme complex formation.
B. Free enzyme ready to bind with another substrate.
C. Release of products.
D. Chemical bonds of the substrate broken
E. Substrate binding to active site.
Choose the correct answer from the options given below:

(1) E, A, D, C, B

(2) A, E, B, D, C

(3) B, A, C, D, E

(4) E, D, C, B, A

63) How many of the following statement about proteins are correct?
(A) They are homopolymers of amino acids.
(B) Collagen is most abundant protein in whole of biosphere.
(C) A peptide bond is formed between carboxyl group of one amino acid and amino group of the next.
(D) The left end of a protein chain is represented by the first amino acid (N-terminal).

(1) Four

(2) One

(3) Two

(4) Three

64) Match **List-I** with **List-II**.

List-I	**List-II**

A.	Primary structure of protein	I.	Human haemoglobin
B.	Secondary structure of protein	II.	Disulphide bonds
C.	Tertiary structure of protein	III.	Polypeptide chain
D.	Quaternary structure of protein	IV.	Alpha helix and β pleated sheet

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-II, C-I, D -IV

(3) A-I, B-III, C-II, D-IV

(4) A-IV, B-III, C-II, D-I

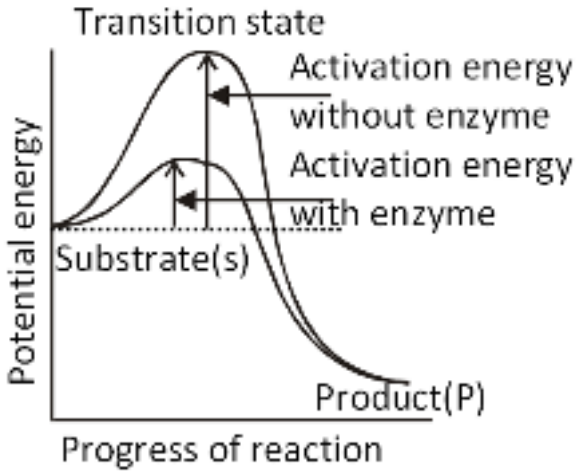
65) At a specific pH, an amino acid exists as a zwitterion, which statement accurately describes this molecular state?

(1) Only the R-group of the amino acid is ionized

(2) The molecule is positively charged due to excess H^+ ions

(3) The molecule carries both positive and negative charges, resulting in a net neutral charge

(4) The amino acid is completely uncharged and non-polar.



66)

Above diagram showing the concept of activation energy. If in this diagram energy level of 'P' become higher than the energy level of 'S' then which of the following will not apply for this diagram ?

(1) The process will be endothermic

(2) Activation energy with enzyme will be less than activation energy without enzyme

(3) Energy difference between transition state and product will be more than activation energy

(4) The substrate will go through the transition state

67) In the B-DNA double helix model, what is the pitch of the helix and distance between two adjacent base pairs respectively?

(1) 3.4 Å and 34Å

(2) 34Å and 3.4Å

(3) 36 nm and 3.6 nm

(4) 36Å and 3.6Å

68) Which amino acid is classified as basic based on nature of its R group?

(1) Valine

(2) Glutamic acid

(3) Lysine

(4) Tyrosine

69) How many of the following are correctly matched with their category of secondary metabolite?
(A) Pigments - Carotenoids, Anthocyanins
(B) Alkaloids - Morphine, Codeine
(C) Toxin - Curcumin, Ricin
(D) Lectin - Concanavalin A
(E) Drugs - Vinblastin, Abrin

(1) Two

(2) Four

(3) Five

(4) Three

70) Match column I and column II :-

	Column I		**Column II**
(A)	Adenine	(i)	Pyrimidine found only in RNA
(B)	Thymine	(ii)	Purine found in both DNA and RNA
(C)	Uracil	(iii)	Pyrimidine also known as 5-methyl uracil
(D)	Cytosine	(iv)	Common pyrimidine in both DNA and RNA

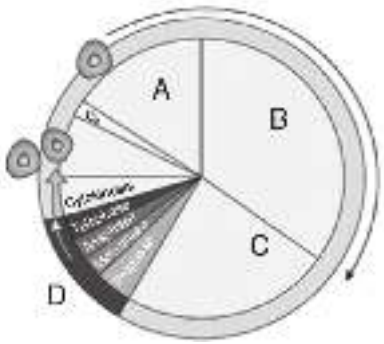
(1) A-ii, B-iii, C-i, D-iv

(2) A-iv, B-iii, C-i, D-ii

(3) A-ii, B-i, C-iii, D-iv

(4) A-iv, B-i, C-iii, D-ii

71) Identify A, B, C and D in the below diagram:



(1) A–G_1, B–S, C–G_2, D–M Phase

(2) A–G_2, B–M Phase, C–G_1, D–S

(3) A–S, B–G_2, C–G_1, D–M Phase

(4) A–M Phase, B–G_1, C–G_1, D–S

72) S-phase is not characterized by -

(1) Chromosome duplication

(2) No increase in chromosome number

(3) DNA replication

(4) Duplication of centriole in nucleus of eukaryotic animal cell

73) If a cell has 2n number of chromosome in G_1 phase, what is the number of chromosome in cell after S-phase?

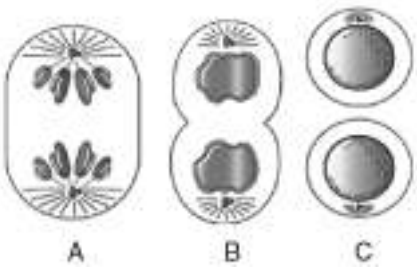
(1) n

(2) 4n

(3) 2n

(4) 8n

74) Identify A, B and C in the below diagram.



(1) A–Interphase, B–Telophase, C–Anaphase

(2) A–Anaphase, B–Telophase, C–Cytokinesis

(3) A–Telophase, B–Cytokinesis, C–Anaphase

(4) A–Cytokinesis, B–Anaphase, C–Telophase

75) Cells at the end of prophase, when viewed under the microscope, do not show -

(1) Golgi body and ER

(2) Nucleolus

(3) Nuclear envelope
(4) All of these

76) Which of the following initiates the start of metaphase?

(1) Completion of bivalent chromosome formation
(2) Assemblage of microtubules of nucleoplasm
(3) Complete disintegration of nuclear envelope
(4) Duplication of chromosome

77) Furrow formation does not occur in plant cell during cytokinesis because of -

(1) Extensible cell wall
(2) Inextensible cell wall
(3) Extensible plasma membrane
(4) Inextensible plasma membrane

78) Which of the following holds true about meiosis?
I. It ensures the production of haploid phase in the life cycle of sexually reproducing organism where fertilization restores the diploid phase.
II. It involves the two sequential cycle of nuclear and cell division called meiosis I and II but only a single cycle of DNA replication.
III. It involves the pairing of homologous chromosomes and recombination between them.
IV. Four haploid cells are formed at the end of meiosis.

(1) I, II, IV only
(2) IV only
(3) I and III only
(4) All of these

79) Select the correct statement from the following:

(1) In leptotene stage the chromosomes become gradually visible under light microscope
(2) During zygotene the heterologous chromosome shows pairing
(3) Chiasmata is a J-shape structure formed in diplotene
(4) Pachytene is characterized by the formation of synaptonemal complex.

80) Homologous chromosomes gets separate during

(1) Metaphase-I
(2) Anaphase-I
(3) Anaphase-II
(4) Telophase-I

81) Diploid somatic cells divides by -

(1) Meiosis

(2) Mitosis only

(3) Both meiosis and mitosis

(4) None of these

82) The number of chromosomes present in pollen grains is six. What shall be their number in leaf cells ?

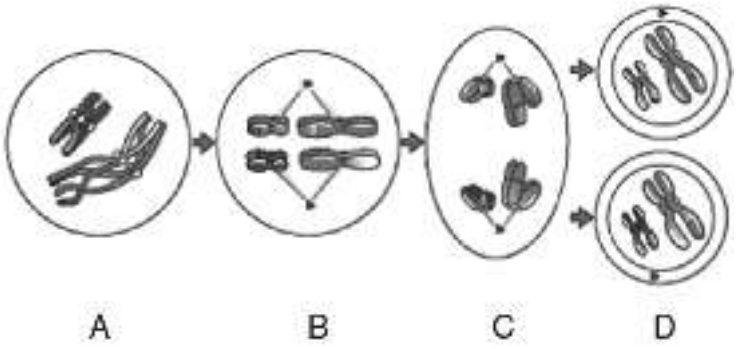
(1) 12

(2) 24

(3) 6

(4) 3

83) The following diagram shows modification of the meiosis I. Identify A, B, C, D.



(1) A–Telophase I, B–Anaphase I, C–Metaphase I, D–Prophase I

(2) A–Prophase I, B–Metaphase I, C–Anaphase I, D–Telophase I

(3) A–Metaphase I, B–Telophase I, C–Prophase I, D–Anaphase I

(4) A–Anaphase I, B–Prophase I, C–Telophase I, D–Metaphase I

84) Meiosis involves :

(1) Two sequential cycles of nuclear and cell divisions and two cycle of DNA replication.

(2) Two sequential cycles of nuclear and cell divisions and one cycle of DNA replication.

(3) One sequential cycles of nuclear and cell division and one cycle of DNA replication.

(4) One sequential cycles of nuclear and cell division and two cycle of DNA replication.

85) Which of the following represents the correct order in Prophase I?

(1) Zygotene, diplotene, pachytene, leptotene, diakinesis

(2) Diakinesis, diplotene, leptotene, pachytene, zygotene

(3) Leptotene, zygotene, pachytene, diplotene, diakinesis

(4) Pachytene, leptotene, zygotene, diplotene, diakinesis

86) Which is the correct statements from the following:
I. Synapsis of homologous chromosomes takes place during prophase I of meiosis.
II. Division of centromeres takes place during anaphase I of meiosis.
III. Spindle fibres disappear completely in telophase of mitosis.

IV. Nucleoli reappear at telophase I of meiosis.

(1) I only

(2) III only

(3) I and II only

(4) I, III and IV only

87) A stage of cell division is shown in the below figure. Select the answer which gives the correct identification of this stage with its characteristics.



(1)	Telophase	Nuclear envelope reforms, Golgi complex reforms
(2)	Late anaphase	Chromosomes move away from the equatorial plate, Golgi complex is not present.
(3)	Cytokinesis	Cell plate formed and mitochondria is distributed between two daughter cells.
(4)	Telophase	Endoplasmic reticulum and nucleolus is not reformed yet.

(1) 1

(2) 2

(3) 3

(4) 4

88) Select the correct option.

	Column I		**Column II**
A.	Synapsis aligns homologous chromosomes	1.	Pachytene
B.	Synthesis of RNA and protein	2.	Zygotene
C.	Action of enzyme recombinase	3.	G_1 phase
D.	Centromeres do not separate but chromosomes move towards opposite poles	4.	Anaphase I

(1) A : 2, B : 1, C : 3, D : 4

(2) A : 2, B : 3, C : 1, D : 4

(3) A : 1, B : 2, C : 3, D : 4
(4) A : 2, B : 3, C : 4, D : 1

89) A bivalent of meiosis-I consists of -

(1) Two chromatids and one centromere
(2) Two chromatids and two centromeres
(3) Four chromatids and two centromeres
(4) Four chromatids and four centromeres.

90) Select the correct statement about G_1 phase.

(1) Cell is metabolically inactive.
(2) DNA in the cell does not replicate.
(3) It is not a phase of synthesis of macromolecules.
(4) Cell stops growing.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	4	1	4	1	2	3	3	1	2	3	3	2	1	3	3	2	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	4	1	1	2	1	4	3	3	3	1	4	3	4	2	4	2	3	1
Q.	41	42	43	44	45															
A.	2	3	3	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	4	1	2	1	4	2	1	1	1	3	4	3	2	2	3	4	3	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	1	2	3	4	4	2	4	3	1	2	1	2	3	3	2	1	3	1	2
Q.	86	87	88	89	90															
A.	1	4	3	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	2	2	3	4	2	3	2	4	3	1	3	2	2	2	1	2	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	1	4	2	3	3	3	1	2	3	2	4	1	3	4	4	4	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	2	3	3	4	2	3	2	4	2	2	2	3	1	4	2	3	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	1	3	1	3	3	2	3	4	1	1	4	3	2	4	3	2	4	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	1	2	2	3	4	1	2	3	2										

SOLUTIONS

PHYSICS

1) **Explaining :** We have to find the maximum speed of particle.
Concept : This question is based on $F = -\nabla U$ and equilibrium.
Solution :
Potential energy $U = \left(\frac{x^2}{2} - x\right)$
For minimum U, $\frac{dU}{dx} = \frac{2x}{2} - 1 = 0$ & $\frac{d^2U}{dx^2} = 1$ = positive
So at x = 1, U is minimum. Hence $U_{min} = -\frac{1}{2}$ J
Total mechanical energy = Max KE + Min PE
Max KE $= \frac{1}{2}mv^2_{max} = 2 - \left(-\frac{1}{2}\right) = \frac{5}{2}$
$\Rightarrow v_{max} = \sqrt{\frac{2}{1} \times \frac{5}{2}} = \sqrt{5}$ ms^{-1}.
Final Answer : (1)

2) **Question Explanation:** The question asks to find the work done by a variable force when a particle moves from x = –a to x = +2a.

Concept: Work done by variable force.

Solution: $F = \frac{b}{a}x$

$$W = \int \vec{F} \cdot d\vec{x}$$

$$W = \frac{b}{a}\int_{x=-a}^{x=0} x\,dx + \frac{b}{a}\int_{x=0}^{x=2a} x\,dx$$

$$W = \frac{b}{2a}\left[x^2\right]_{x=-a}^{x=0} + \frac{b}{2a}\left[x^2\right]_{x=0}^{x=2a}$$

$$W = -\frac{ab}{2} + 2ab = \frac{3ab}{2}$$

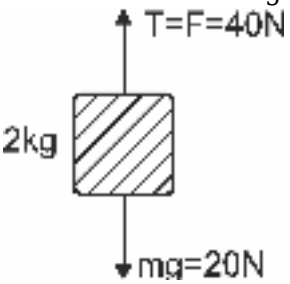
Final answer: Option (1)

3) **Explaining :** Find the work done on the body in first 2 sec.
Concept : This question is based on work-energy theorem.
Solution :
Slope (a) $= \frac{0-50}{10-0} = -5$ m/s^2
By the graph at t = 0 sec, $v_1 = 50$ m/s
velocity at t = 2 sec, $v_2 = 50 - (5)(2) = 40$ m/s
work done is change in kE

$\omega = \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2} \times 10 \times (1600 - 2500) = -4500J$

Final Answer : (4)

4) **Explaining :** Find the work done by tension on the block.
Concept : This question is based on work done by tension.
Solution :
Acceleration of block is = 10 m/s^2
$\therefore$ displacement s = $\frac{1}{2}$ at^2 = $\frac{1}{2}$ × 10 × 4 = 20m
Tension in the string is 40 N



and work done by tension is
ω = T× s = 40 × 20 = 800 J
Final Answer : (1)

5) **Explaining :** Find the maximum compression in spring.
Concept : This question is based on conservation of mechanical energy.
Solution :
From COME
$\frac{1}{2}kx^2 = mg(x + h)$
$k = 1960\frac{N}{m},\ h = 0.4\,m,\ m = 2kg$
g = 9.8 ms^2
$\Rightarrow \frac{1}{2}$ (2) (1960)^2 = 2 × 9.8 (x + 0.4)
⇒ x = 0.10 m
Final Answer : (4)

6) **Explaining :** Find the total work done.
Concept : This question is based on work done by constant force.
Solution :
Work done = Area covered in between force displacement curve and displacement axis
= Mass × Area covered in between acceleration–displacement curve and displacement axis.
$= 10 \times \frac{1}{2}(8 \times 10^{-2} \times 20 \times 10^{-2})$
$= 8 \times 10^{-2}J$
Final Answer : (1)

7) **Explaining :** Find the total work done on the particle.
Concept : This question is based on work done by variable force.
Solution :

For (0, 0) to (0, 2)

$$W_1 = \int_0^0 2xdx + \int_0^2 3ydy$$

$$= 0 + \left[\frac{3y^2}{2}\right]_0^2 = 6J$$

For (0, 2) to (2, 2)

$$W_2 = \int_0^2 2xdx + \int_2^2 3ydy$$

$$= \left[x^2\right]_0^2 + 0 = 4J$$

Total work = $W_1 + W_2$

= 6 + 4 = 10 J

Final Answer : (2)

8) **Explaining :** Find the K.E. of block at position C.

Concept : This question is based on conservation of mechanical energy.

Solution :

Gain in KE = Loss in PE

$$\frac{1}{2}mv^2 - 0 = mg(h_1 - h_2)$$

$$\frac{1}{2}mv^2 = (1)\,(10)\,(14-7)$$

= 70 J

Final Answer : (3)

9) **Question Explanation:** Find pump power factor to multiply water mass flow rate by 3.

Concept: Power and Mass Flow Rate.

Solution: As we know that

$$P = \frac{1}{2}\left(\frac{dm}{dt}\right)V^2 \quad \& \quad \frac{dm}{dt} = PAV$$

So, that $P \propto \left(\frac{dm}{dt}\right)^3$

therefore to mape $\frac{dm}{dt} \rightarrow$ 3 times $\rightarrow 3\left(\frac{dm}{dt}\right)$

the, $P_{new} = (3)^3 P_{old}$

= 27 × 2

= 54 hP

10) **Explanation Question:**

Asking about distance covered by a car before doubling the speed, if power delivering to the car is constant.

Concept:

This question is based on Power & use of calculus

Solution:
P = Fv
 F = ma
 $a = \frac{vdv}{dx}$
$P = mv^2\frac{dv}{dx}$
$\int_0^x \frac{P}{m}dx = \int_V^{2V} v^2 dv$
$\frac{P}{m}x = \left[\frac{v^3}{3}\right]_v^{2v}$
$\frac{P}{m}x = \frac{1}{3}\left[(2v)^3 - (v)^3\right]$
$\frac{P}{m}x = \frac{8v^3 - v^3}{3}$
$x = \frac{m}{P}.\frac{7v^3}{3}$
$x = \frac{7mv^3}{3P}$

Final Answer :
Option (1)

11)

Question Explanation: Find momentum percentage change for a given KE percentage increase.

Concept: This question is based on kinetic energy & linear momentum.

Solution: As we know, that
$P = \sqrt{2mK}$
i.e. $P \propto \sqrt{K}$
then % change in the momentum will be given as

$$\left(\frac{P_2}{P_1} - 1\right) \times 100 = \left(\sqrt{\frac{KE_2}{KE_1}} - 1\right) \times 100$$
$$= \left(\sqrt{\frac{500}{100}} - 1\right) \times 100 = 123\%$$

Final answer: The correct option is (2).

12)

Question Explanation: Identifying properties of conservative forces.

Concept: Conservative Forces and Work Done.

Solution: A force is called conservative if the work it does on an object depends only on where the object starts and ends, not on the path taken. Because the path doesn't matter, if an object returns to its starting point (a closed loop), the total work done by that force is always zero. While many conservative forces (like gravity) are central, it is not a requirement. Potential energy changes as work is done, so it does not remain constant.

A. Statement (i): True. By definition, work depends only on initial and final positions.

B. Statement (iii): True. Since work is path-independent, $w_{AB} = -w_{BA}$, making the total work for a round trip zero.

Final Answer: Option 3 (i, iii)

13) **Question Explanation:** Evaluate assertion and reason about conservative and non-conservative forces.

Concept: This question is based on conservative and non-conservative forces.

Solution: Assertion: Work done by conservative forces does not depend on path taken. (True, this is the definition).
Reason: Work done by non-conservative forces depends only on initial and final position. (False, work done by non-conservative forces depends on the path taken).

Final answer: The correct option is (3).

14) **Explaining :** Find the work done by given forces and match the correct values with column-II.
Concept : This question is based on work done by variable force.
Solution :
(P) $W = mgh = 2 \times 10 \times 2 = 40\ J \rightarrow$ (ii)
(Q) $W = Fd\cos 60^\circ = 10 \times 4 \times 0.5 = 20\ J \rightarrow$ (iii)
(R) Gravity and Normal force are perpendicular to displacement. $W = 0\ J \rightarrow$ (i)
(S) $W = (mg\sin\theta) \times d = (2 \times 10 \times \sin 30^\circ) \times 5 = 10 \times 5 = 50\ J \rightarrow$ (iv)
Final Answer : (2)

15) **Explanation the Question :** Find spring constant from max compression on a rough surface.
Concept: This question is based on work-energy theorem.
Solution:
$w_{friction} = \Delta KE + \Delta U$

$-\mu mg(\square + x) = +\frac{1}{2}mu^2 + \frac{1}{2}kx^2$

$-0.1 \times 50 \times (10.5) = \frac{1}{2}(5)(20)^2 + \frac{1}{2}K(0.5)^2$

$\frac{1}{8}K = 1000 - 52.5 = 947.5$

$\boxed{K = 7,580\ N/m}$

Final answer : (1)

16) **Explaining :** To find the current through the cylinder.
Concept : This question is based on non uniform current density.
Solution :

$$I = \int_R J.ds$$

$$= \int_0 J_0 \left(1 - \frac{x}{R}\right).2\pi x dx$$

$$= 2\pi J_0 \int_0^R \left(x - \frac{x^2}{R}\right) dx$$

$$= 2\pi J_0 \left(\frac{x^2}{2} - \frac{x^3}{3R}\right)_0^R$$

$$I = 2\pi J_0 \left(\frac{R^2}{2} - \frac{R^2}{3}\right)$$

$$I = 2\pi J_0 \frac{R^2}{6}$$

$$I = \frac{\pi J_0 R^2}{3}$$

Final Answer : (3)

17) **Explaining :** To choose the current statements.
Concept : This question is based on Drift Velocity
Solution :
Electric current is same in different portions of wire but current density (or electric field) is different in different portions of the wire.

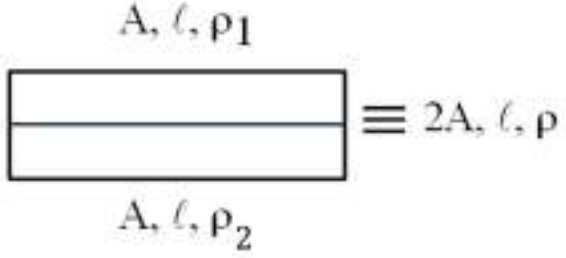
From $I = neAv_d$ we have $v_d \propto \frac{1}{A}$

Final Answer : (3)

18) **Explain the question :** Find effective resistivity of combination.

Concept : This question is based on for parallel combination $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

Solution : In given question two wire are connected in parallel so $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ having same dimensions.



$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{2A}{\rho\ell} = \frac{A}{\rho_1\ell} + \frac{A}{\rho_2\ell}$$
$$\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$$

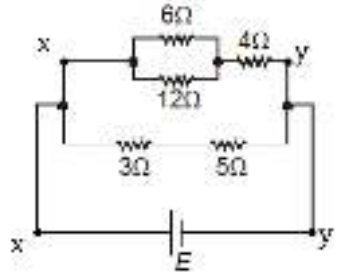
Final answer : Option **(2)**

19) **Explaining :** To fine $R_1 + R_2$ is proportional to
Concept : This question is based on VI characteristics
Solution :
Slope of VI curve = Resistance
$R_1 = \tan\theta$; $R_2 = \tan(90 - \theta) = \cot\theta$

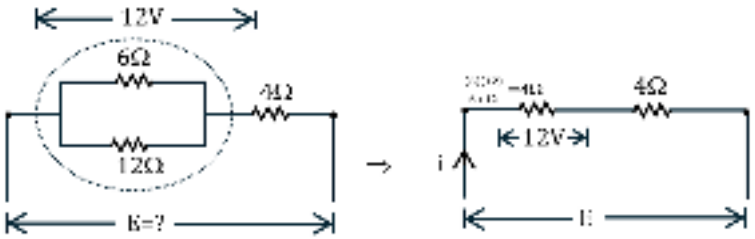
$$R_2 + R_1 \propto \tan\theta + \cot\theta = \tan\theta + \frac{1}{\tan\theta}$$
$$= \frac{\sin\theta + \cos\theta}{\cos\theta \ \sin\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin\theta \ \cos\theta} = \frac{1}{\sin\theta \ \cos\theta}$$

Final Answer : (2)

20) **Explanation :** To find EMF when potential drop is 12 volt across 6Ω as shown in circuit.
Concept : This question is based on KVL
Solution :

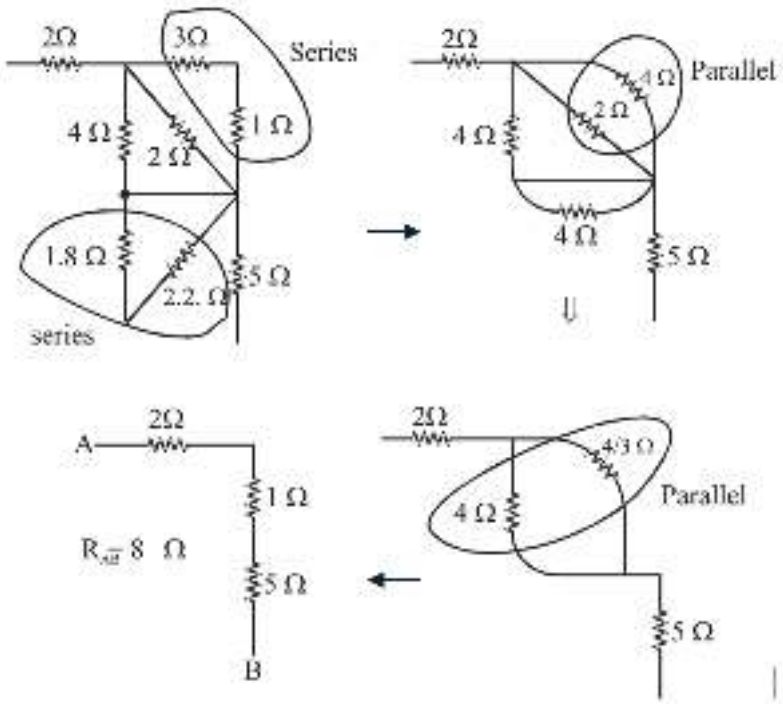


From the above diagram the branch having 6Ω, 12Ω & 4Ω is in parallel with "E" therefore,



E = 12 + 12
= 24 volt
Final Answer : (3)

21) **Explaining :** To find equivalent resistance
Concept : This question is based on combination of resistors
Solution :



Final Answer : (2)

22)

Explain the Question : In this question, we have to find how electric field (E) and current density (J) are related in a conductor.
Concept : This question is based on relation between electric field and current density
Solution :
E = ρJ, where = resistivity.
Hence, E ∝ J (for same material)
Final answer : 2

23) **Explaining :** To find emf of each cell
Concept : This question is based on combination of cells
Solution :
For combination V = 10E –(10r)i
at i = 12 A ; V = 6V ; Put values
6 = 10 E –(10r) (12) .....(i)
at i = 14 A ; V = 0
0 = 10 E –(10r) (14) ⇒ 10 E = 140r .... (ii)
E = 4.2 Volt
Final Answer : (4)

24) **Explaining :** To find value of n & m.
Concept : This question is based on Battery grid
Solution :

$R = 2.5\,\Omega, \quad r = 0.5\,\Omega$

no. of rows = m, no of cells in each row = n

$i = \dfrac{nE}{R + nr/m}$ total cells $\Rightarrow$ mn = 45

for max. current ; $R = nr/m$, $mR = nr$

$n = 5m$, $5m^2 = 45$, $m = 3$, $n = 15$

Final Answer : (1)

25)

Explain the Question : We need to calculate maximum power dissipated in resistance R of the given circuit.

Concept : This question is based on maximum power transfer occurs when $R = r_{eq}$.

Solution :

$E_{eq} = (10 + 5) = 15V$ and

$r_{eq} = (3 + 2) = 5\Omega$

For maximum power in R

$R = r_{eq} = 5\Omega$

So current

$$I = \frac{E_{eq}}{R + r_{eq}} = \frac{15}{5+5} = 1.5A$$

Power in resistance R

$P = I^2R$

$= (1.5)^2 \times 5W$

$= 11.25$ W

Final answer : 1

26) **Explain Question:**

Find which bulb glow with maximum brightness.

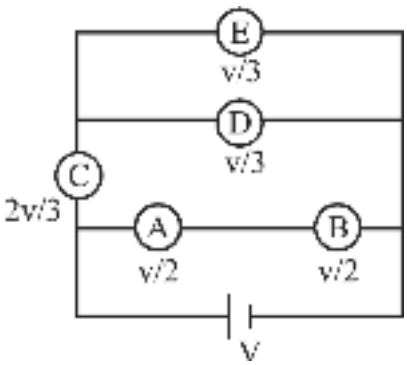
Concept:

Brightness of bulb $(P) = \dfrac{V^2}{R}$.

Solution:

In given question all bulbs are identical i.e. having same resistance.

So $P \times V^2$



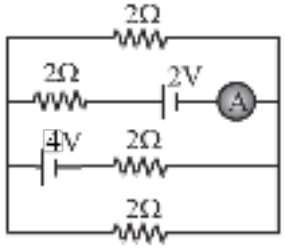
Potential difference of (C) is

Maximum hence bulb (C) lights with maximum brightness.

Final Answer: Option (2)

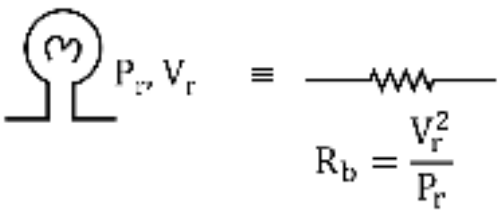
27) **Explaining :** To find value of shunt resistance
Concept : This question is based on Ammeter
Solution :
$(i - i_g)\ S = i_g G$
$(100 - 10)\ S = 10 \times 100$
$S = \frac{1000}{90}$
$= 11.11\,\Omega$
Final Answer : (4)

28) **Explaining :** Given : Electric circuit
Ask : Reading of ammeter
Concept : This question is based on key points :
(i) $\text{KVL} \quad : \quad \Sigma\Delta V = 0 \quad (\text{In loop})$
(ii) $\text{KCL} \quad : \quad \Sigma i = 0 \quad (\text{At junction})$
Solution :
$I_1 = \frac{x-2}{2} = \frac{\Delta V}{R}$
$I_2 = \frac{x-4}{2}$
$I_3 = \frac{x-0}{1}$
x–2 + x –4 + 2x = 0
4x = 6
x = 3/2 = 1.5 V



Let say I_1, I_2 and I_3 are current in branches applying KCl on assuming potential at lift as x and on Right as zero.
$I_1 = \frac{1}{4}A$
Final Answer : (4)

29) **Explaining :** Given : Electric circuit with bulb
Ask : Brightness of bulb
Concept : This question is based on key points :



$R_b = \frac{V_r^2}{P_r}$
(i)
(ii) $\text{Brightness} \propto P_{\text{consumption}}$
Solution :

$P = I^2R$

Calculation :- As all bulls are identical and the power (related by brightness) is given as $P = I^2R$ so me bulb with the most current will be brightest. In Parallel connection, the current is spilt between the branches while in series the current will be same.

So in figure we can set mat bulb 'C' will receive into maximum current while the bulbs A, B, D will not get complete current as 'C' as ***m*** are connected in Parallel ***trenches***. So bulb 'C' will be the brightest.

Conclusion : The current answer option 3.

Final Answer : (3)

30) **Explaining :** Given : Electric network

Ask : Ratio of equivalent resistance of three different cases

Concept : This question is based on key points :

(i) $\text{Series} : R_{eq} = R_1 + R_2$

(ii) $\text{Parallel} : R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

(iii) Circuit reduction

Solution :

$$R_{AB} = r \| \left(\frac{r}{2} + \frac{r}{3}\right) = \frac{5r}{11}$$

$$R_{BC} = \frac{r}{2} \| \left(r + \frac{r}{3}\right) = \frac{4r}{11}$$

$$R_{AC} = \frac{r}{3} \| \left(r + \frac{r}{2}\right) = \frac{3r}{11}$$

5 : 4 : 3

Final Answer : (3)

31) **Explanation :** To find direction of flow of current in galvanometer.

Concept : This question is based on galvanometer (wheat stone bridge)

Solution :

If $\dfrac{R_3}{R_4} > \dfrac{R_1}{R_2}$ Current B → D

Final Answer : (3)

32) **Explaining :** Energy dissipated when 100 μF capacitor charged to 200 V is discharged through resistor.

Concept : This question is based on All stored energy converts to heat.

Solution :

C = 100μf V = 200V

Heat generated = $\dfrac{1}{2} CV^2$

$$\frac{1}{2} \times 100 \times 10^{-6} \times 200 \times 200 = 2J$$

Final Answer : (1)

33) **Explaining :** Energy stored in parallel plate capacitor with field (E) between plates separated by (d), area (A).

Concept : This question is based on energy density

Solution :

$$U = \frac{1}{2}CV^2$$

$$U = \frac{1}{2}\left(\frac{A\in_0}{d}\right)(Ed)^2 = \frac{1}{2}A\in_0 E^2 d$$

Final Answer : (4)

34) **Explain Question:**
A dielectric slab is inserted between the plates of a charged condenser (isolated capacitor). We have to determine the change in capacitance, charge, potential difference, and energy stored.

Concept:
For an isolated charged capacitor, charge remains constant; inserting a dielectric increases capacitance, hence potential difference and energy decrease.

Solution:

A. On inserting a dielectric, capacitance increases:

C' = kC

A. Since the condenser is charged and isolated, charge remains unchanged:

Q' = Q

A. Potential difference:

$$V = \frac{Q}{C} \Rightarrow V' = \frac{Q}{kC} = \frac{V}{k}$$

Hence, potential difference decreases.

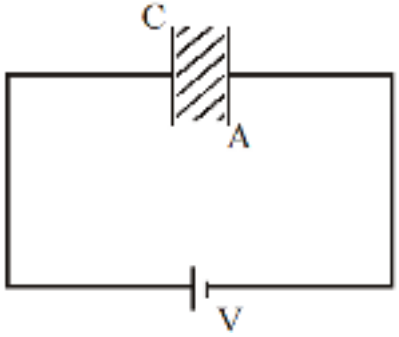
A. Energy stored:

$$U = \frac{Q^2}{2C} \Rightarrow U' = \frac{Q^2}{2kC} = \frac{U}{k}$$

Hence, energy stored decreases.
Therefore:
Capacitance increases, charge remains unchanged, potential difference decreases, and energy stored decreases.

Final answer: Option (3)

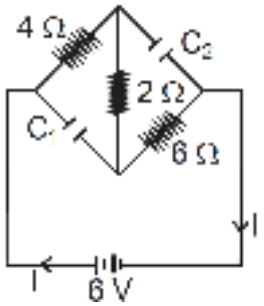
35) **Explanation :** To find the dielectric constant is the capacitance
Concept : This question is based on breakdown voltage
Solution :



$A = 10^{-4}\ m^2$
$E_{max} = 10^6\ V/m$
$C = 15\ \mu F$
$$C = \frac{k\varepsilon_0 A}{d}$$
$$\frac{Cd}{\varepsilon_0 A} = k$$
$$k = \frac{15 \times 10^{-12} \times 500 \times 10^{-6}}{8.86 \times 10^{-12} \times 10^4}$$
$$= \frac{15 \times 5}{8.86} = 8.465$$
$k \approx 8.5$
Final Answer : (4)

36) **Explaining :** To solve for energy lost in redistribution of charge
Concept : This question is based on Energy lost, $\Delta U = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V^2)$
Solution :
$$\Delta U = \frac{1}{2}\frac{(600\mu)(600\mu)}{(1200\mu)}(4 \times 10^4) = 6J$$
Final Answer : (2)

37) **Explaining the Question:** To find the ratio of charge stored in C_1 and C_2.
Concept : Charge on capacitor is Q = CV
Solution :



In steady state
Req = 12Ω
$$I = \frac{6}{12} = 0.5A$$
P.D across C_1 = 3V
P.D across C_2 = 4V
$q_1 = C_1V_1 = 12\ \mu C$
$q_2 = C_2V_2 = 24\ \mu C$
$$\frac{q_1}{q_2} = \frac{1}{2}$$
Final Answer : (4)

38) **Explaining the Question:** To choose correct graph between i and t.
Concept : Q = CV
$$\frac{dQ}{dt} = V\frac{dC}{dt}$$

$i = V\frac{dC}{dt}$

Solution :

Q = CV

C↑ , Q↑ ,

$i = \frac{dQ}{dt} = V\frac{dC}{dt}$

$\frac{dC}{dt}$ = constant,

i ∝ Constant

Final Answer : (2)

39)

Question Explanation:

Five parallel metal plates of area A are arranged equidistantly at d. Plates 1 and 4 are connected, and plates 3 and 5 are connected. The capacitance between A and B is required.

Concept :

Capacitance (combination of plotes)

Formula Used:

• Capacitance of a parallel plate capacitor:

$C = \frac{\varepsilon_o A}{d}$

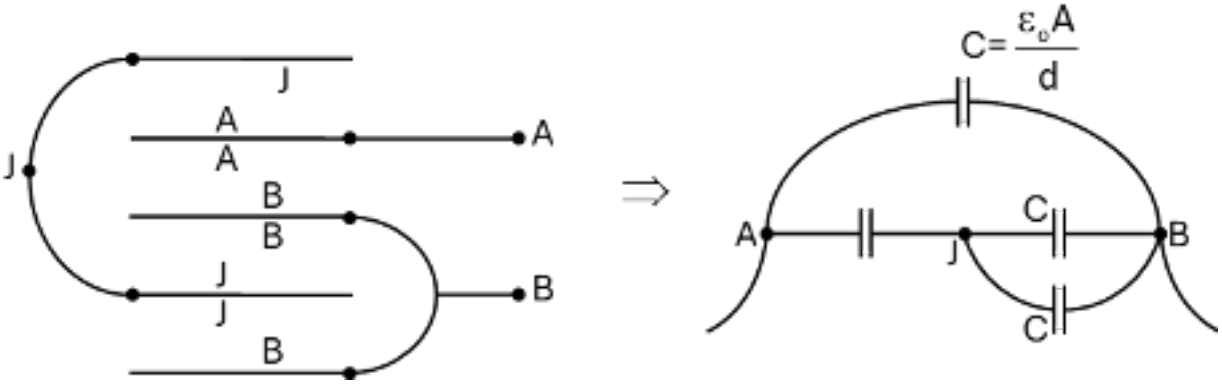
• For series capacitors:

$\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$

• For parallel capacitors:

$C_{eq} = \sum C_i$

Visual Diagram:



Calculation:

$C_{eq} = C + \left(\frac{(C)(2C)}{C+2C}\right)$

$C_{eq} = C + \frac{2}{3}C$

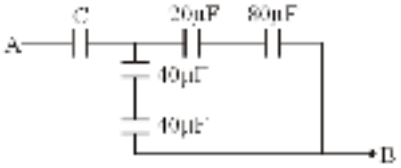
$C_{eq} = \frac{5}{3}C$ ⇒ $C = \frac{\varepsilon_o A}{d}$

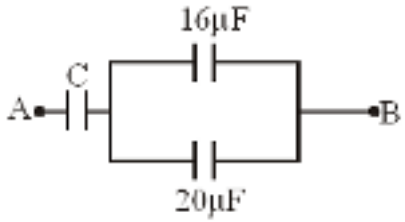
$$\boxed{C_{eq} = \frac{5}{3}\frac{\varepsilon_0 A}{d}}$$

Final Answer:

The correct option is (3).

40) **Explanation :** To find equivalent capacitance between A and B
Concept : This question is based on equivalent capacitance
Solution :
Equivalent diagram



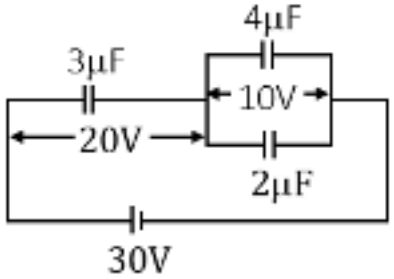


≡

$$\equiv \frac{C \times 36}{C + 36} = 12$$

C = 18 μF
Final Answer : (1)

41) **Explanation:** For the circuit shown, match the charge supplied by the battery and the charges on specific capacitors.
Concept: Capacitors in series have the same charge, and capacitors in parallel have the same voltage. The total charge is $Q = C_{eff} V$.
Solution:



(i) $Q_{battery} = C_{eff}\, V$
$= 2 \times 30 = 60\ \mu C$
(ii) $Q_{4\mu F} = 4 \times 10$
$= 40\ \mu C$
(iii) $Q_{2\mu F} = 2 \times 10$
$= 20\ \mu C$
(iv) $Q_{3\mu F} = 3 \times 20$
$= 60\ \mu C$
Final Answer : (2)

42) **Explanation :** To find charge on the given capacitor
Concept : This question is based on RC circuit
Solution :
Current in 2Ω is 1 amp.
Potential difference across capacitor is 2 volt
Q = CV = 10 μC
Final Answer : (3)

43) **Explanation :** To find the smallest of the three resistance
Concept : This question is based on RC circuit
Solution :
Slope $\frac{dq}{dt} = i = i_0 e^{-t/\tau}$
At t =0, Slope = $i^0 = \frac{V}{R} = \frac{Q}{CR}$
Q Slope 1 > Slope2 > Slope3
 $R_1 < R_2 < R_3$
Final Answer : (3)

44) **Explain the Question :** In the given RC circuit, we need to find the current through 2Ω resistor in steady state.
Concept : This question is based on RC circuit at steady state.
Solution :
At steady state no current is flow in capacitor branch
$I = \frac{6}{\frac{2\times3}{2+3} + 2.8} = \frac{6}{4} = \frac{3}{2}$ and $I_{2\Omega} = \frac{3}{2+3} \times I$
$= \frac{3}{5} \times \frac{3}{2}$
= 0.9 A
Final Answer : 2

45) **Explanation :** To find the potential difference across each of the capacitor
Concept : This question is based on sharing of charges
Solution :
$V_{Common} = \frac{Q_1 + Q_2}{C_1 + C_2}$
$= \frac{CV + 2CV}{C + 2KC} = \frac{3V}{2K + 1}$
Final Answer : (4)

CHEMISTRY

46)

Explaining : Calculate EMF of cell.
Concept : This question is based on $E^o_{cell} = (SRP)_C - (SRP)_A$
Solution :

$E^{o}_{cell} = E^{o}_{Ag^{+}|Ag} - E^{o}_{Cu^{2+}|Cu}$

$= +0.8\ V - 0.3V = 0.5\ V$

Final Answer : (2)

47)

Explaining : Calculate E^{o}_{cell}

Concept : This question is based on $E^{o}_{cell} = (SRP)_C - (SRP)_A$

Solution : $E^{o}_{cell} = -0.41 - 0.15 = -0.56\ V$

Final Answer : (2)

48)

Explaining : Calculate $E_{cell.}$

Concept : This question is based on nernst equation

Solution : Net cell reaction :

$2Cr + 3Fe^{2+} \rightarrow 2Cr^{3+} + 3Fe$

$E_{cell} = +0.72 - 0.42 - \frac{0.059}{6}\log\left\{\frac{(0.1)^2}{(0.01)^3}\right\} = +0.26V$

Final Answer : (4)

49)

Explaining : Find reaction quotient.

Concept : This question is based on Nernst equation

Solution : $E = E^{o} - \frac{2.303RT}{nF}\log_{10}Q$

$\because E = E^{\circ} \Rightarrow Q = 1$

Final Answer : (1)

50)

Explaining : Calculate equilibrium constant.

Concept : This question is based on Nernst equation.

Solution :

$0.2905 = E^{\circ} - \frac{0.059}{2}\log\left(\frac{0.01}{0.001}\right)$; $E° = 0.32\ V$

$\Rightarrow E^{\circ} = 0.32 = +\frac{0.059}{2}\log k$

$\Rightarrow K = 10^{\frac{+0.32}{0.0295}}$

Final Answer : (2)

51)

Explain the Question : To find molar conductivity of acetic acid (HOAc) at infinite dilution using known values of other electrolytes.

Concept : This question is based on By Kohlrausch's Law of Independent Migration of Ions,

Λ^0(HOAc) = Λ^0(HCl) + Λ^0(NaOAc) – Λ^0(NaCl).
Solution :
Given Λ^0(NaOAc) = 91.0 S cm^2 mol^{-1},
Λ^0(HCl) = 426.2 S cm^2 mol^{-1}.
Hence, to calculate Λ^0(HOAc), we also need Λ^0(NaCl).
Final Answer : (1)

52)

Explaining : Calculate number of faradays.
Concept : This question is based on apply faradays law
Solution :
$MnO_4^- + 5e \longrightarrow Mn^{2+}$
1 mole 5 mole ; No. of E = No. of Faradays
5 mole e^- = 5 Faradays mol × V_f = No. of F
1 × 5 = No. of F
Final Answer : (4)

53)

Explaining : Calculate mass of iron deposited.
Concept : This question is based on apply faradays law.
Solution :
No. of F = No. of Faradays
$\frac{w}{56/2} = 3$; w = 84 g
Final Answer : (2)

54)

Explaining : Find number of electrons.
Concept : This question is based on Faradays law
Solution :
$\frac{5 \times 200}{96500} \times 6.023 \times 10^{23} = 6.24 \times 10^{21}$
{No. of Faradays = No. of mol of e^-}
$\frac{I \times t}{96500}$ = mol of e^-
Final Answer : (1)

55)

Explaining : Calculate solubility.
Concept : This question is based on application of Kohlrausch law.
Solution :
$\lambda^o_{AgCl} = \lambda^o_{Ag^+} + \lambda^o_{Cl^-} = 130$; $\Lambda^\infty_m = \frac{k \times 1000}{S_{mol/L}}$ for sparingly soluble salt.
$130 = 3.4 \times 10^{-6} \times \frac{1000}{S}$

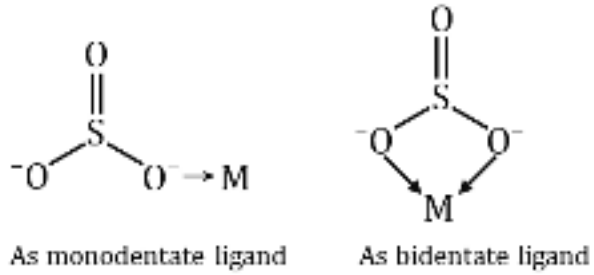
$\Rightarrow S = 2.6 \times 10^{-5}$
Final Answer : (1)

56)

Explaining : - The common features among the species CN^-, CO and NO^+ are :
Concept : This question is based on types of ligand.
Solution :
CN^-, CO and NO^+ have bond order three and diamagnetic.
Final Answer : (1)

57)

Explaining : - Which of the following ambidentate ligand can act as flexidentate ligand ?
Concept : This question is based on types of ligand.
Solution :
SCN^-, OCN^-, SO_3^{2-} all are ambidentate ligand, but only SO_3^{2-} can act as flexidentate ligand.



Final Answer : (3)

58)

Explaining : - CO bond length is longest in which complex?
Concept : This question is based on metal carbonyl compounds.
Solution :
$(\text{Bond length})_{CO} \propto \frac{1}{(\text{Bond order})_{CO}}$; All central have same configuration but Ti^{-2} has strongest M–C back bond so bond length C–O is longest.

$(\text{Bond order})_{CO} \propto \frac{1}{\text{Back donating tendency to metal atom/ion}}$
Final Answer : (4)

59)

Explaining : - What is the oxidation number of Fe in $[Fe(H_2O)_5(NO)]^{2+}$ ion?
Concept : This question is based on oxidation state calculation.
Solution :
$[Fe^{+1}(H_2O)_5(NO)^{2+}]$ ion
Final Answer : (3)

60)

Explaining : - Incorrect match for given complex compound/ion and its characteristics :
Concept : This question is based on isomerism in complex compound.
Solution : $[CoBr_3(H_2O)_3]$: It does not exhibit hydrate isomerism.
Final Answer : (2)

61)

Explaining : - Find IUPAC name of the hydrate isomer of $CrCl_3.6H_2O$, which is having lowest electrical conductivity.
Concept : This question is based on IUPAC
Solution :
$[Cr(H_2O)_4Cl_2]Cl.2H_2O$; hydrate isomer of complex $[Cr(H_2O)_6]Cl_3$ has lowest electrical conductivity, because it gives only two ions on ionization in aqueous medium.
Final Answer : (2)

62)

Explaining : - If EAN of central metal cation M^{2+} in a non-chelating complex is 36 and atomic no. of metal M is 26, then the number of monodentate ligand in this complex are :
Concept : This question is based on EAN rule.
Solution :
EAN = Z–O.S. + 2 × C.N. ; 36 = 26 – 2 + 2x
x = 6 = Co-ordination no.
Final Answer : (3)

63)

Explaining : Effective atomic number
Concept : This question is based on Effective atomic number
Solution :
EAN = Z–O.S. + 2 × C.N.

	$[Fe(CN)_6]^{3-}$ and	$[Fe(NH_3)_6]^{3+}$
EAN	35	35
	$[Cr(NH_3)_6]^{3+}$ and	$[Cr(CN)_6]^{3-}$
EAN	33	33
	$[FeF_6]^{3-}$ and	$[Fe(CN)_6]^{3-}$
EAN	35	35
	$[Ni(CO)_4]$ and	$[Ni(CN)_4]^{2-}$
EAN	36	34

Final Answer : (4)

64)

Explaining : IUPAC Nomenclature.
Concept : This question is based on IUPAC Nomenclature.
Solution :
$[Fe(O_2)(CN)_4Cl]^{4-}$ ion
IUPAC name – chlorotetracyano superoxo ferrate (II) ion.

Final Answer : (3)

65)

Explaining : - Select the correct IUPAC name of Prussian blue :
Concept : This question is based on IUPAC.
Solution :
Prussian blue : $Fe_4[Fe(CN)_6]_3$ or $Fe^{3+}[Fe^{2+}(CN)_6]^{4-}$
Iron(III) hexacyanoferrate(II)
Final Answer : (3)

66)

Explaining : - Which of the following complex compound is “Pentaaquacyanidoiron (III)trichloridotricyanido cobaltate (III)” ?
Concept : This question is based on IUPAC.
Solution :
In “Pentaaquacyanidoiron (III)trichloridotricyanido cobaltate (III)”.
Complex cation is $[Fe^{III}(CN)(H_2O)_5]^{2+}$ and complex anion$[Co^{III}Cl_3(CN)_3]^{3-}$
Therefore complex is $[Fe(CN)(H_2O)_5]_3[CoCl_3(CN)_3]_2$
Final Answer : (4)

67)

Explaining : No. of ions.
Concept : This question is based on Werner's Theory
Solution :
$[Pt(NH_3)_5Cl]Cl_3$ produces four ions per molecule
$[Pt(NH_3)_6Cl]Cl_4$ produces five ions per molecule
$[Pt(NH_3)_2Cl_4]$ produces 0 ions per molecule
$[Pt(NH_3)_4Cl_2]Cl_2$ produces three ions per molecule in the solution.
Final Answer : (1)

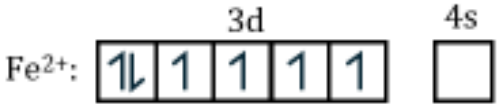
68)

Explaining : - Iron carbonyl, $Fe(CO)_5$ is :
Concept : This question is based on metal carbonyl.
Solution :
Based on the number of metal atoms present in a complex, they are classified as :
e.g., : $Fe(CO)_5$: mononuclear
$Co_2(CO)_8$: dinuclear
$Fe_3(CO)_{12}$: trinuclear
Final Answer : (2)

69)

Explaining : Crystal field theory.
Concept : This question is based on Crystal field theory.

Solution :
In $K_4[Fe(CN)_6]$ is complex, Fe is in +2 oxidation state.



As CN^- is a strong field ligand, it causes pairing of electrons therefore, electronic configuration of Fe^{2+} in $K_4[Fe(CN)_6]$ is $t_{2g}^6 e_g^0$.
Final Answer : (3)

70)

Explaining : - Which of the following is the correct order of increasing field strength of ligands to form coordination compounds ?
Concept : This question is based on spectrochemical series.
Solution :
Spectrochemical series (as given in NCERT) :

$$I^- < Br^- < SCN^{\ominus} < Cl^- < S^{2-} < F^- < \bar{O}H < C_2O_4^{2-} < H_2O < NCS^{\ominus} < EDTA^{4-} < NH_3 < en < \overset{\ominus}{C}N < CO$$

Final Answer : (4)

71)

Explaining : Magnetic moment
Concept : This question is based on Magnetic moment
Solution :
(A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)
Formula : $\sqrt{n(n+2)}B.M.$
where n = number of unpaired electron.
Final Answer : (4)

72)

Explaining : - Applications of Crystal field theory
Concept : This question is based on Applications of Crystal field theory
Solution :
Ti^{4+} and Sc^{3+} have d^0 configuration whereas Zn^{2+} has d^{10} configuration. These are colourless species. Cr^{3+} has d^3 configuration. Due to d-d transitions, it is coloured.
Final Answer : (2)

73)

Explaining : - Which one of the following has an optical isomer ?
Concept : This question is based on isomerism in complex compound.
Solution :
Read in optical isomerism of type $[M(AA)_3]$. Here AA is a symmetrical didentate ligand.
Final Answer : (4)

74)

Explaining : - C.F.S.E.
Concept : This question is based on Crystal field theory
Solution :

In$[Cr(CN)_6]^{3-}$ complex ion, $Cr^{3+} = d^3 = t_{2g}^3$

In$[Co(NH_3)_6]^{3+}$ complex ion, $Co^{3+} = d^6 = t_{2g}^6$ (low spin)

In$[CoCl_6]^{3-}$ complex ion, $Co^{3+} = d^6 = t_{2g}^4 e_g^2$; Weaker the ligand lesser the splitting

In$[Co(H_2O)_6]^{3+}$ complex ion, $Co^{3+} = d^6 = t_{2g}^6$

Final Answer : (3)

75)

Explaining : Cis platin formula.
Concept : This question is based on application of coordination compounds.
Solution : The complex cis-platin, $[Pt(NH_3)_2Cl_2]$ is anti-cancer.
Final Answer : (1)

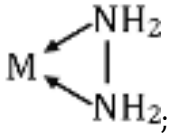
76)

Explaining : Types of Ligand.
Concept : This question is based on Synergic Bonding.
Solution :

CO is a π-acid ligand $M \underset{\pi}{\overset{\sigma}{\rightleftharpoons}} L$.
Final Answer : (2)

77)

Explaining : - Statement based question.
Concept : This question is based on types of ligand.
Solution :
N_2H_4 can only act as monodentate ligand.



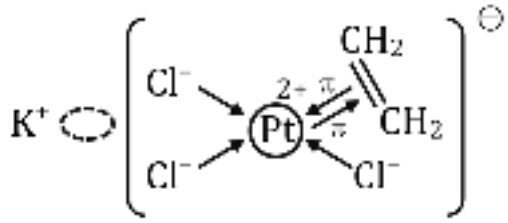
; Three membered (unstable) ring. Hence it acts as monodentate ligand only.
Final Answer : (1)

78)

Explaining : Color in coordination compounds.
Concept : This question is based on applications of coordination chemistry.
Solution :
Both $[Ti(H_2O)_6]^{4+}$ and $[Sc(H_2O)_6]^{3+}$ are colourless, due to absence of free electrons in 3d subshell.
Final Answer : (2)

79)

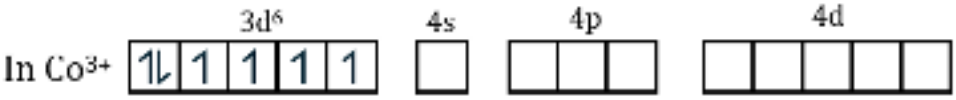
Explaining : - Statement based question.
Concept : This question is based on organometallic compound.
Solution :
Zeise's salt is a π-bonded organo-metallic compound, $K[PtCl_3(\pi\text{-}C_2H_4)]$
Zeise's salt contains C_2H_4 molecule as one of the ligand.



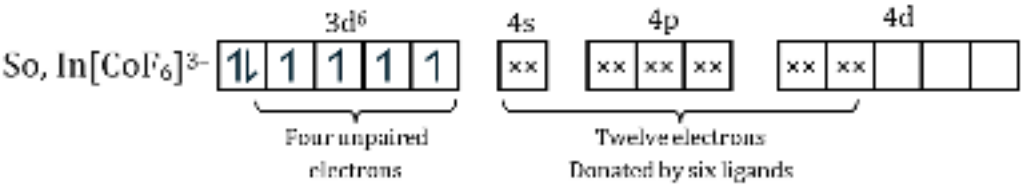
Final Answer : (3)

80)

Explaining : Crystal field theory.
Concept : This question is based on Crystal field theory.
Solution :
In complex $[CoF_6]^{3-}$,
Oxidation number of Co = +3
Electronic configuration of
$Co^{3+} = 1s^2,2s^22p^6, 3s^23p^63d^6$



In Co^{3+} number of unpaired electrons is four, and F^- is the ligand of weak field. Hence, it does not pair up the electrons of d-orbital.



So, $In[CoF_6]^{3-}$ is paramagnetic
Co^{3+} has $3d^6$ outer electronic configuration, but four unpaired electrons do not pair up because of weak field provided by F^-.
Final Answer : (3)

81) **Explaining :** Find best reducing agent ?
Concept : This question is based on application of ECS.
Solution : Lower the reduction potential, higher is the reducing power.
Final Answer : (2)

82)

Explaining : Products at anode ?
Concept : This question is based on electrolysis
Solution :

Reaction at anode (oxidation) :
$2Cl^- \rightarrow Cl_2(g) + 2e^-$
Final Answer : (1)

83) **Explaining :** Amount of electricity ?
Concept : This question is based on electrolysis
Solution : $Cu^{2+} + 2e^- \rightarrow Cu(s)$
1 mole copper requires 2 mole electron, i.e., 2 faraday charge.
Final Answer : (3)

84)

Explaining : $\Lambda_{eq} = ?$
Concept : This question is based on electrolytic conductance
Solution :
$\kappa = G \times G^*$
$\kappa = \frac{1}{210} \times 0.66 = 0.0031428\ \Omega^{-1}\ cm^{-1}$
$\Lambda_{eq} = \frac{\kappa \times 1000}{N} = \frac{0.0031428 \times 1000}{0.01}$
$\Lambda_{eq} = 314.28\ \Omega^{-1}\ cm^2\ eq^{-1}$
Final Answer : (1)

85)

Explaining : Find cell constant = ?
Concept : This question is based on electrolytic conductance
Solution : $\kappa = G \times G^*$
$G^* = \frac{\kappa}{G} = \frac{0.0035 \times 500}{1} \quad \left(\because G = \frac{1}{R}\right)$
$G^* = 1.75\ cm^{-1}$
Final Answer : (2)

86)

Explaining : Find ΔG° = ?
Concept : This question is based on relation between ΔG° and E°
Solution : $\Delta G^\circ = \frac{-6 \times 96500 \times 2.02}{1000} = -1170 kJ$
Final Answer : (1)

87)

Explain the question: Find ratio of cations reduced when equal moles of electrons are passed through $AgNO_3$, $CuSO_4$, and $AuCl_3$ solutions.
Concept: Amount of metal deposited ∝ moles of electrons / n-factor.
Solution:

For reduction:
$Ag^{+} + e^{-} \rightarrow Ag$ (n = 1)
$Cu^{2+} + 2e^{-} \rightarrow Cu$ (n = 2)
$Au^{3+} + 3e^{-} \rightarrow Au$ (n = 3)
Moles of electrons = 3 mol each.
Moles of cations reduced = (moles of e^{-}) / n
$Ag^{+} \rightarrow 3/1 = 3$
$Cu^{2+} \rightarrow 3/2 = 1.5$
$Au^{3+} \rightarrow 3/3 = 1$
Ratio = 3 : 1.5 : 1 = 6 : 3 : 2
Final Answer: 4

88) **Explaining :** Find out strong and weak electrolyte ?
Concept : This question is based on graph between Λ_m and $\sqrt{C}$.
Solution :
A → Represents strong electrolyte (KCl)
B → Represents weak electrolyte (CH_3COOH)
Final Answer : (3)

89) **Explaining :** Relation between Λ_m and Λ_{eq}
Concept : This question is based on electrolytic conductance
Solution :
$Fe_2(SO_4)_3 \rightarrow 2Fe^{+3} + 3\ SO_4^{2-}$
n–factor = 6
$\Lambda_{eq} = \frac{\Lambda_m}{6}$
Final Answer : (2)

90) **Explaining :** Properties at salt bridge solution
Concept : This question is based on salt bridge
Solution : Velocity of both K^{+} and NO_3^{-} are nearly the same
Final Answer : (1)

BIOLOGY

91) **Solution:** Each testicular lobule contains **1–3 highly coiled seminiferous tubules**. Seminiferous tubules contain **Sertoli cells and male germ cells**, while Leydig cells occur in the **interstitial spaces** and secrete androgens.
Final Answer : (1)

92) **Solution:** The correct sequence is: Spermatogonia → Primary spermatocyte → Secondary spermatocyte → Spermatid → Spermatozoa
Primary spermatocyte undergoes **meiosis I**, while secondary spermatocytes undergo **meiosis II**.
Final Answer : (1)

93) **Solution:** Meiosis I separates homologous chromosomes but sister chromatids remain together.
Therefore: Secondary spermatocyte = 23 chromosomes + 46 chromatids.
Final Answer : (2)

94) **Solution: Spermiogenesis** is the process by which **spermatids transform into spermatozoa**. Release of spermatozoa into the lumen is called **spermiation**.
Final Answer : (2)

95) **Solution: LH** acts on Leydig cells and stimulates them to secrete **androgens, mainly testosterone**.
Final Answer : (3)

96) **Solution:** Testosterone is secreted by **Leydig cells**, whereas Sertoli cells provide nourishment/support to developing germ cells and secrete **inhibin**.
Final Answer : (4)

97) **Solution:** Oogonia multiply during fetal life and differentiate into **primary oocytes**, which remain arrested in **prophase I** until puberty.
Final Answer : (2)

98) **Solution:** Ovulation generally occurs around **day 14** and is triggered by the **LH surge**.
Final Answer : (3)

99) **Solution:** A sharp **LH surge** around the middle of the menstrual cycle causes **rupture of the mature Graafian follicle and ovulation**.
Final Answer : (2)

100) **Solution:** During the **follicular phase**, ovarian follicles develop under the influence of FSH. Developing follicles secrete increasing amounts of estrogen.
Final Answer : (4)

101) **Solution:** After ovulation, the ruptured Graafian follicle forms the **corpus luteum**, which mainly secretes **progesterone**.
Final Answer : (3)

102) **Solution:** If pregnancy does not occur, the corpus luteum degenerates, causing a fall in **progesterone**, which contributes to breakdown of the endometrium and menstruation.
Final Answer : (1)

103) **Solution:** Fertilisation generally occurs at the **ampullary-isthmic junction** of the fallopian tube.
Final Answer : (3)

104) **Solution:** The **acrosome** contains enzymes that help sperm penetrate the **zona pellucida** surrounding the secondary oocyte.
Final Answer : (2)

105) **Solution:** Fusion triggers the **cortical reaction**, which modifies the zona pellucida and prevents **polyspermy**.
Final Answer : (2)

106) **Solution:** Ovulation releases a **secondary oocyte**, which is arrested at **metaphase II**. Meiosis II is completed only after sperm entry.
Final Answer : (2)

107) **Solution:** Following fertilisation: Zygote → Cleavage → Morula → Blastocyst → Implantation
Final Answer : (1)

108) **Solution:** The blastocyst becomes attached to and gets embedded in the **endometrium**, the inner lining of the uterus.
Final Answer : (2)

109) **Solution:** The **trophoblast** forms the outer layer of the blastocyst and contributes to implantation and **placenta formation**.
Final Answer : (2)

110) **Solution: Human chorionic gonadotropin (hCG)** is secreted by the trophoblast/placental tissue and can be detected in maternal blood and urine during pregnancy.
Final Answer : (3)

111) **Solution:** Placenta produces hormones such as **hCG, hPL, estrogens and progesterone**. **FSH** is a pituitary gonadotropin.
Final Answer : (4)

112) **Solution: hCG maintains the corpus luteum**, allowing continued secretion of progesterone during early pregnancy.
Final Answer : (2)

113) **Solution:** Parturition involves complex neuroendocrine mechanisms. **Oxytocin** promotes powerful uterine contractions, while cortisol promoting fetal organ maturation.
Final Answer : (1)

114) **Solution: Prolactin** stimulates milk production by mammary glands, whereas **oxytocin** causes contraction of myoepithelial cells, resulting in **milk ejection**.

Final Answer: (4)

115) **Solution: A — Correct: LH → Leydig cells → androgens**
B — Correct: FSH → Sertoli cells → supports spermatogenesis
C — Correct: Fertilisation occurs at the ampullary-isthmic junction
D — Incorrect: Implantation occurs in the **endometrium**, not myometrium
E — Correct: hCG maintains the corpus luteum during early pregnancy
Final Answer: (2)

116) **Solution:** Combined oral contraceptive pills contain **estrogen and progesterone/progestin**. They primarily inhibit ovulation and also alter cervical mucus, making it difficult for sperm to enter the uterus.
Final Answer: (3)

117) **Solution:** Copper-releasing IUDs such as **Cu-T** increase **phagocytosis of sperms** within the uterus. Copper ions also suppress sperm motility and fertilising capacity.
Final Answer : (3)

118) **Solution: LNG-20** is a hormone-releasing IUD. It releases levonorgestrel and helps prevent pregnancy.
Final Answer : (3)

119) **Solution:** Condoms provide a **barrier** that reduces the exchange of semen, vaginal secretions and infectious agents. Thus, they help prevent pregnancy as well as many STIs.
Final Answer : (1)

120) **Solution: Vasectomy** involves cutting and tying the **vas deferens**, preventing sperm from entering the ejaculate.
Final Answer : (2)

121) **Solution:** In **tubectomy**, the fallopian tubes are cut and tied, preventing the meeting/transport of sperm and ovum.
Final Answer : (3)

122) **Solution:** Emergency contraceptive pills are recommended **within 72 hours** of unprotected intercourse for effective prevention of pregnancy. Their effectiveness generally decreases with delay.
Final Answer : (2)

123) **Solution: Cu-T is an intrauterine device**, whereas periodic abstinence, withdrawal and lactational amenorrhea are natural methods.
Final Answer : (4)

124) **Solution:** Intense and prolonged breastfeeding can suppress ovarian activity and

ovulation. The method is effective only under specific conditions and for a limited postpartum period.
Final Answer : (1)

125) **Solution:** Gonorrhoea, syphilis and genital herpes are STIs. **Malaria** is transmitted primarily through the bite of infected female *Anopheles* mosquitoes.
Final Answer : (3)

126) **Solution:** AIDS is caused by **HIV (Human Immunodeficiency Virus)**, not *Plasmodium*.
Final Answer : (4)

127) **Solution:** STIs can cause **genital itching, abnormal discharge, painful urination, sores and swelling**, although some infections may remain asymptomatic.
Final Answer : (4)

128) **Solution:** Sharing needles can transmit blood-borne infections such as **HIV and hepatitis B** and should be strictly avoided.
Final Answer : (4)

129) **Solution: MTP is relatively safer during the early stages of pregnancy**, particularly during the first trimester, compared with later stages.
Final Answer : (1)

130) **Solution:** In **IVF**, fertilisation occurs outside the body and the resulting embryo is transferred into the female reproductive tract, making it useful in cases where fallopian tubes are blocked.
Final Answer : (1)

131) **Solution: GIFT (Gamete Intra-Fallopian Transfer)** involves transferring **gametes** into the fallopian tube so that fertilisation can occur inside the female reproductive tract.
Final Answer: (2)

132) **Solution: ICSI (Intracytoplasmic Sperm Injection)** involves direct injection of a sperm into the cytoplasm of the ovum.
Final Answer: (3)

133) **Solution: ZIFT (Zygote Intra-Fallopian Transfer)** involves transferring a **zygote or early embryo** into the fallopian tube.
Final Answer : (2)

134) **Solution:** A-Q, B-P, C-R, D-S
GIFT → Gametes → fallopian tube
ICSI → Sperm directly injected into ovum
ZIFT → Zygote/early embryo → fallopian tube

IVF-ET → Embryo → female reproductive tract
Final Answer: (3)

135) **Solution :** A, B, D and E only
A — Correct
B — Correct
C — Incorrect — Vasectomy blocks the **vas deferens**; it does not stop sperm formation.
D — Correct
E — Correct
Final Answer: (3)

136) **Explanation :** Iodine test is used to detect starch.
Amylose (a component of starch) reacts with iodine to give a blue or violet-blue color.
Cellulose does not react with iodine because it is a linear polymer and does not contain helices so cannot hold iodine molecules. So, the statement is incorrect (mis-match).
Final Answer : (4)

137)

Explanation:

Biomacromolecules are large molecules with high molecular weight, typically polymers made up of repeating monomeric units. Proteins, polysaccharides, and nucleic acids are true biomacromolecules because they are formed by polymerization.

A. Lipids, however, are not strictly biomacromolecules because they do not form true polymers. Instead, they are composed of smaller units like fatty acids and glycerol but are not linked in long chains like proteins or nucleic acids.

   Correct Answer: 2. Lipids

138)

Explanation:
The secondary structure of a protein refers to the localized folding of the polypeptide chain, stabilized by hydrogen bonds. The alpha (α) helix is one such common structure, where the backbone coils in a right-handed spiral due to intramolecular hydrogen bonding between the carbonyl (C=O) of one amino acid and the amide (N-H) of another, four residues away.

Correct Answer: 3. Secondary structure of protein

139) **Solution:**
Full form of NAD is Nicotinamide Adenine Dinucleotide.
NAD is a coenzyme found in all living cells. It plays a crucial rote in redox reactions (reduction-oxidation reactions), which are involved in energy metabolism. There are two forms of NAD:

A. NAD^+ (oxidized from): This form accepts electrons and is reduced to NADH during metabolic reactions.
B. NADH (reduced form): NADH donates electrons in other reactions, such as those in the electron transport chain during cellular respiration.

Final Answer : (2)

140) **Solution:**
Change In substrate Concentration: With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first. The reaction ultimately reaches a maximum velocity (Vmax). This velocity is not exceeded by any further rise in concentration of the substrate.
Reason: The enzyme molecules are fewer than the substrate molecules and after saturation of these enzyme molecules, there are no free enzyme molecules available to bind with the additional substrate molecules.

Hence,
Correct option: (4) First increases then become constant

141)

Solution :
The Michaelis–Menten constant (Km) is defined as the substrate concentration at which the velocity of an enzyme-catalysed reaction is half of its maximum velocity (Vmax/2).
Km is a characteristic constant for a particular enzyme–substrate system.

Other options like half-life, concentration ratio, and turnover number refer to different enzyme properties.

Final answer : (2)

142)

Solution: Starch is a polysaccharide of glucose. Its helical structure can hold iodine molecules, producing a characteristic blue colour with iodine solution. Cellulose is also a glucose polymer but does not form this iodine-positive helical structure.
Final answer: (2)

143) **Solution :** Proteins function as biocatalyst responsible for increasing rate of reaction. Carbohydrates yield energy because they are stored food.
Final Answer : (2)

144)

Question Explanation: Serine has a hydroxymethyl ($–CH_2OH$) R group.

Concept: Amino Acid R Groups

Solution:

A. Glycine → H

B. Alanine → Methyl (–CH_3)

C. Serine → Hydroxymethyl (–CH_2OH)

D. Valine → Isopropyl (–$CH(CH_3)_2$)

Therefore, Serine – Hydroxymethyl group is correctly matched.

Final answer: Option - 3 (Serine – Hydroxymethyl group)

145)

Solution: Adenylic acid is nucleotide formed by adenine, ribose sugar and phosphate group; it is also called adenosine monophosphate (AMP).
Final answer: (1)

146)

Solution:
Some enzymes have allosteric site to control active site. This control is called modulation.
• Most of non-competitive inhibitors blind reversibly with allosteric site and negatively change the configuration of active site. Such inhibition is called non-competitive reversible inhibition or negative allosteric modulation.

Hence,
Correct option: (4) Both activation and inhibition

147)

Question Explanation: A product of a metabolic pathway can inhibit an earlier enzyme.

Concept: Feedback Inhibition

Solution: When the end product of a biochemical pathway inhibits an enzyme involved in an earlier step, it is called feedback inhibition. This prevents unnecessary production of the end product and helps regulate the metabolic pathway.

Final answer: Option - 2 (Feedback inhibition)

148)

Question Explanation: In competitive inhibition, substrate and inhibitor compete for the active site.

Concept: Enzyme Inhibition

Solution: In competitive inhibition, the inhibitor resembles the substrate and competes with it for the active site of the enzyme. Increasing substrate concentration can overcome competitive inhibition. Non-competitive inhibition cannot normally be overcome by increasing substrate concentration.

Final answer: (3)

149) **Solution :-** Prosthetic groups are tightly bound to the aponzyme (permenent association).
Coenzymes are loosely bound and can detach after the reaction.
Final Answer : (2)

150) **Explanation :**

A. (a) is incorrect because lipids with only single bonds are called **saturated fatty acids**, not unsaturated.
B. (b) is **correct** : Lecithin is a type of **phospholipid**.
C. (c) is correct : Glycerol is a trihydroxy propane because it has three hydroxyl groups (–OH)
D. (d) is incorrect because **palmitic acid** has **16 carbon atoms**, not 20.
E. (e) is incorrect because **arachidonic acid** has **20 carbon atoms**, not 16.

151) **Solution :**
(a) Lecithin is a phospholipid.
(b) Saturated fatty acid does not possess double bond between carbon atoms.
Final Answer : (2)

152) **Question Explanation :** Identifies the correct sequence of the enzyme catalytic cycle.
Concept : The enzyme catalytic cycle.
Solution :

The correct sequential steps for the catalytic cycle of an enzyme action are: E. Substrate binding to active site → A. Substrate enzyme complex formation → D. Chemical bonds of the substrate broken → C. Release of products → B. Free enzyme ready to bind with another substrate.

Final Answer : (1)

153) **Solution :** Statement A and B are incorrect. Proteins are heteropolymers of amino acid, not homopolymers. RuBisCO is the most abundant protein in the biosphere.
Final Answer : (3)

154) **Solution :** A–III, B–IV, C–II, D–I

A.	Primary structure of protein	III.	Polypeptide chain
B.	Secondary structure of protein	IV.	Alpha helix and β pleated sheet
C.	Tertiary structure of protein	II.	Disulphide bonds
D.	Quaternary structure of protein	I.	Human haemoglobin

Final Answer : (1)

155) **Solution :** The amino group is NH^{+}_{3} and the carboxyl group is COO^{-}, creating a dipolar ions.
Final Answer : (3)

156)

Solution:
Since the diagram represents activation energy, it likely shows the energy changes during a chemical reaction. Here, the key change is that P (Product) has a higher energy than S (Substrate), meaning the reaction absorbs energy, making it endothermic.
Correct Answer: Option 3: "Energy difference between transition state and product will be more than activation energy" is incorrect.
Activation energy (Ea) is the energy difference between the transition state and substrate, not between the transition state and product.
The energy difference between the transition state and product is usually smaller than activation energy.

157) **Solution :** One full turn (pitch) is 34Å long and contains 10 base pairs, making the rise per base pair 3.4Å.
Final Answer : (2)

158) **Solution :** Lysine, Arginine and histidine are classified as basic amino acid.
Final Answer : (3)

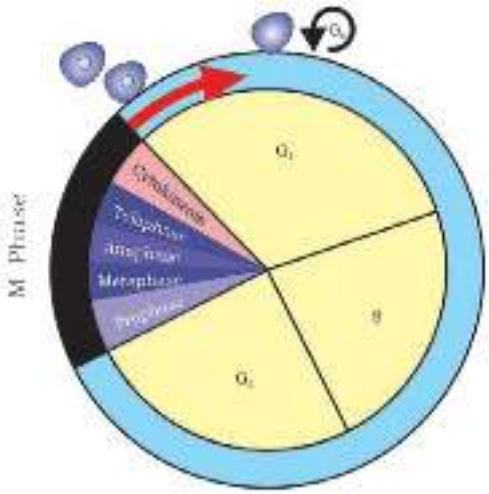
159) **Solution :** C and E are incorrectly matched. Toxins are Abrin and Ricin. While drugs are Vinblastin and Curcumin
Final Answer : (4)

160) **Solution :**

	Column I		**Column II**
(A)	Adenine	(ii)	Purine found in both DNA and RNA
(B)	Thymine	(iii)	Pyrimidine also known as 5-methyl uracil
(C)	Uracil	(i)	Pyrimidine found only in RNA
(D)	Cytosine	(iv)	Common pyrimidine in both DNA and RNA

Final Answer : (1)

161) **Solution :** A–G_1, B–S, C–G_2, D–M Phase



Final Answer : (1)

162) **Solution :** S-phase is not characterized by duplication of centriole in nucleus of eukaryotic animal cell.
Final Answer : (4)

163) **Solution :** The number of chromosomes in cell after S-phase are 2n as in S-phase amount of DNA per cell doubles but chromosome number remains same.
Final Answer : (3)

164) **Solution :** A–Anaphase, B–Telophase, C–Cytokinesis
Final Answer : (2)

165) **Solution :** Cells at the end of prophase, when viewed under the microscope, do not show Golgi body, ER, Nucleolus and Nuclear envelope.
Final Answer : (4)

166) **Solution :** -
Final Answer : (3)

167) **Solution :** Furrow formation does not occur in plant cell during cytokinesis because of inextensible cell wall.
Final Answer : (2)

168) **Solution :** All statements are correct.
Final Answer : (4)

169) **Solution :** During zygotene, the homologous chromosome shows pairing. Chiasmata is a X-shape structure formed in diplotene. Pachytene is characterized by recombination.
Final Answer : (1)

170) **Solution :** Homologous chromosomes gets separate during anaphase-I.

Final Answer : (2)

171) **Solution :** Diploid somatic cells divides by mitosis only.
Final Answer : (2)

172) **Solution :** Leaf has diploid cell i.e. 2n = 12.
Pollen grains haploid cell n = 6.
Final Answer : (1)

173) **Solution : A–**Prophase I, **B–**Metaphase I, **C–**Anaphase I, **D–**Telophase I.
Final Answer : (2)

174) **Solution :** Meiosis involves two sequential cycles of nuclear and cell divisions and one single cycle of DNA replication.
Final Answer : (2)

175) **Solution :** Leptotene, zygotene, pachytene, diplotene, diakinesis.
Final Answer : (3)

176) **Solution :** Division of centromeres takes place during anaphase II of meiosis.
Homologous chromosomes separate during anaphase I.
Final Answer : (4)

177) **Solution :** Telophase - Nuclear envelope reforms, Golgi complex reforms.
Final Answer : (1)

178) **Solution :** A : 2, B : 3, C : 1, D : 4

	Column I		**Column II**
A.	Synapsis aligns homologous chromosomes	2.	Zygotene
B.	Synthesis of RNA and protein	3.	G_1 phase
C.	Action of enzyme recombinase	1.	Pachytene
D.	Centromeres do not separate but chromosomes move towards opposite poles	4.	Anaphase I

Final Answer : (2)

179) **Solution :** A bivalent of meiosis-I consists of four chromatids and two centromeres because bivalent consist of two homologous chromosomes.
Final Answer : (3)

180) **Solution :** Cell is metabolically active, macromolecules like RNA and proteins are synthesized and cell grow maximally.

Final Answer : (2)